

The Ministry of Civil Aviation (Height Restrictions for Safeguarding of Aircraft Operations) Rules, 2015

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The Ministry of Civil Aviation (Height Restrictions for Safeguarding of Aircraft Operations) Rules, 2015

UNION OF INDIA

India

The Ministry of Civil Aviation (Height Restrictions for Safeguarding of Aircraft Operations) Rules, 2015

Rule

THE-MINISTRY-OF-CIVIL-AVIATION-HEIGHT-RESTRICTIONS-FOR-SAFEGUARDING-OF-AIRCRAFT-OPERATIONS-RULES-2015 of 2015

Published on 30 September 2015

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The Ministry of Civil Aviation (Height Restrictions for Safeguarding of Aircraft Operations) Rules, 2015

Published vide Notification No. G.S.R. 751(E), dated 30th September, 2015

Ministry of Civil Aviation

G.S.R. 751 (E). - The Central Government, being of opinion that it is necessary and expedient to do so for the safety of aircraft operations, proposes to make the following certain rules, in exercise of the powers conferred by sub-section (1) and clause (o) and clause (r) of sub-section (2) of section 5 read with section 9A of the Aircraft Act, 1934 (XXII of 1934) (hereinafter referred to as the said Act), and in supersession of the Ministry of Civil Aviation notification number

S.O.84(E), dated the 14th January, 2010 published in the Gazette of India, Part II, section 3, sub-section (ii), except as respect things done or omitted to be done before such supersession. The objections or suggestions on the draft S.O. were called from the stakeholders and are considered by the Government to the extent admissible. In the public interest the rules are notified by seeking exemption from putting the rules again in the public domain.

1.

Short title and commencement.

(1)

These rules may be called the Ministry of Civil Aviation (Height Restrictions for Safeguarding of Aircraft Operations) Rules, 2015.

(2)

They shall come into force on the date of their publication in the Official Gazette.

2.

Application .

- These rules shall apply to all civil and defence aerodromes listed in Schedule III to Schedule VII, as amended from time to time.

3.

Definitions.

- In these rules, unless the context otherwise requires, -

(i)

"Act" means the Aircraft Act, 1934 (XXII of 1934);

(ii)

"aerodrome" includes civil and defence airports, airstrips, communication, navigation and surveillance facilities used for the aeronautical purposes in India;

(iii)

"aerodrome elevation" means the elevation of the highest point of the landing area as specified in Schedule III to Schedule VII;

(iv)

"Authorised officer" means the officer authorised by the defence authorities for the purposes of these rules;

(v)

"Airports Authority" means the Airports Authority of India constituted under section 3 of the Airports Authority of India Act, 1994 (55 of 1994

);

(vi)

"Colour Coded Zoning Map" of an airport means the map, prepared and certified by the Airports Authority indicating the areas around the airport in different colour coded grids with the permissible heights above mean sea level for the purpose of aerodrome safeguarding and a model Colour Coded Zoning Map of Navi Mumbai International Airport is given at Schedule IX;

(vii)

"Designated officer" means the officer of Airports Authority of India responsible for processing and/or issuance of No Objection Certificate as specified in Appendix M of Schedule VIII;

(viii)

"No Objection Certificate" means the certificate issued under rule 5;

(ix)

"structure" includes building, mast, tower, chimney, poles, transmission lines, elevated roads or viaducts or bridges and elevated railway lines, wind farms and all other man-made structures;

(x)

Words and expressions used but not defined in these rules shall have the meanings respectively assigned to them in the Act.

4.

Restrictions on constructions, erections, trees, etc.

(1)

No structure shall be constructed or erected, or any tree planted or grown on any land within a radius not exceeding twenty kilometers from the Aerodrome Reference Point of the civil and defence aerodromes, as specified in Schedule III to Schedule VII, without obtaining a No Objection Certificate for the height clearance, except in cases specified in sub-rule (2) of rule 7.

(2)

No structure shall be constructed or erected, or any tree planted or grown on any land within the areas specified in Schedule I of the civil and defence aerodromes, as listed in Schedule III to Schedule VII, except for essential navigational aids and other installations required for aeronautical purposes.

(3)

No structure higher than the height specified in Schedule II, shall be constructed or erected and no tree, which is likely to grow or ordinarily grows higher than the height specified in the said Schedule shall be planted on any land within a radius of twenty kilometers from the Aerodrome Reference Point.

(4)

The level roads and level railway lines within one kilometer of the airport boundary wall shall also be subject to issuance of the No Objection Certificate.

5.

Issuance of No Objection Certificate.

(1)

The No Objection Certificate in respect of civil aerodromes shall be issued by the designated officer on behalf of the Central Government in respect of civil aerodromes.

(2)

The No Objection Certificate in respect of defence aerodromes shall be issued by the authorised officer in accordance with Schedule I and Schedule II, subject to such other conditions as the said authorised officer may deem fit.

(3)

In case of State owned and private aerodromes, licensed by the Directorate General of Civil Aviation, the No Objection Certificate for the protection of obstacle limitation surfaces (OLS) at such airports shall be issued by the designated officer and the procedure in cases of State owned and private aerodromes, not licensed by the Directorate General of Civil Aviation, shall be regulated in the manner as specified in rule 13.

(4)

The application for issuance of No Objection Certificate in respect of civil aerodromes, shall be made by the applicant to the designated officer through the No Objection Certificate Application System (NOCAS), accessible on the website of the Airports Authority at www.aai.aero.

6.

Issuance of Colour Coded Zoning Map.

- The Colour Coded Zoning Maps (CCZM) shall be issued by the Airports Authority based on the latitude & longitude of the area in respect of civil aerodromes which shall indicate through different colour coded grids, the permissible heights in the areas around the airport, falling within the radius not exceeding twenty kilometers from the Aerodrome Reference Point. The CCZM shall be available at AAI website www.aai.aero.

7.

Approval for construction of buildings, structures, etc.

(1)

After considering the No Objection Certificate issued by the designated officer or the authorised officer, the concerned Local, Municipal or Town Planning and Development Authorities shall approve the construction of buildings or structures not exceeding the Permissible Top Elevation. Local, Municipal or Town Planning and Development Authorities shall also consider the existing building regulations or bye-laws or any other law for the time being in force before approving the construction of buildings or structures.

(2)

In cases of aerodromes where the Colour Coded Zoning Maps has been issued, the Local, Municipal or Town Planning and Development authorities shall, in accordance with the height specifications provided in such Colour Coded Zoning Maps, approve the construction of the structures, as per the existing building regulations or bye laws or any other law for

the time being in force:

Provided that no such approval shall be given by the Local, Municipal or Town Planning and Development authorities for sites which lies in approach, take off and transitional areas of an airport or in any other area, marked in the Colour Coded Zoning Map for the compulsory obtaining of No Objection Certificate from the designated officer or authorised officer.

(3)

The Local, Municipal or Town Planning and Development authorities shall certify on the sanction plan that the Floor Space Index or Floor Area Ratio and the related height of the building or structure is within the permissible elevation as indicated in the Colour Coded Zoning Map for the given site.

(4)

The Local, Municipal or Town Planning and Development Authorities shall submit the details of structures approved under sub-rule (1) and sub-rule (2) to the concerned designated officer or the authorised officer within a period of thirty days from the date of such approval.

8.

Clearances for siting towers of fixed wireless stations.

(1)

The clearances in respect of siting towers of fixed wireless stations shall be issued by the Standing Advisory Committee on Radio Frequency Allocation (SACFA) of the Ministry of Communication & Information Technology, Government of India, taking into consideration the heights above mean sea level specified in the Colour Coded Zoning Map.

(2)

The Standing Advisory Committee on Radio Frequency Allocation (SACFA) of the Ministry of Communication, Government of India, shall submit the details of siting towers of fixed wireless stations approved under sub-rule (1) to the concerned designated officer or the authorised officer within a period of thirty days from the date of such approval.

9.

Processing of No Objection Certificate cases.

(1)

Processing of NOC cases in respect of civil aerodromes shall be carried out at nine Airports Authority offices one each at Delhi, Kolkata, Mumbai, Chennai, Guwahati Hyderabad, Bengaluru, Ahmedabad and Nagpur airports.

(2)

The designated officer available at the offices specified in sub-rule (1) shall be responsible for the processing of applications and issue of No Objection Certificate and/or issue authorization for issuance of NOC for height clearance by concerned designated officer with respect to the civil aerodromes.

(3)

A Panel of Chartered Engineers and Surveyors may be assigned by the Airports Authority to carry out physical verification of details of Site Elevation and Coordinates as submitted by the applicant. The expenses of the same shall be borne by the applicant.

(4)

The officer in-charge of the corporate office at the headquarters of the Airports Authority in New Delhi shall supervise the functioning of the regional and station level offices.

10.

Duties of designated officer.

(1)

The designated officers, specified in Appendix M of Schedule VIII, shall be responsible for issuance of No Objection Certificate in respect of civil aerodromes and shall coordinate with the respective Local, Municipal or Town Planning and Development authorities in granting approval for construction of buildings or structures.

(2)

The designated officer shall forward the copy of NOCs issued by him under sub-rule (1) above to the concerned airport operator and respective Local, Municipal or Town Planning & Development authorities.

11.

Appellate Committee.

(1)

There shall be an Appellate Committee consisting of the following, namely:-

- (a)
Joint Secretary (Airports), Ministry of Civil Aviation, Government of India - Chairperson;
- (b)
Joint Director General of Civil Aviation (Aerodrome), Directorate General of Civil Aviation - Member;
- (c)
Member (Air Navigation Services), Airports Authority of India - Member; and
- (d)
One technical expert having knowledge in the field of communication or air traffic management - Member.

(2)
If any person or Local, Municipal or Town Planning and Development authorities or any airport operator is aggrieved with the decision of the Designated officer, such person or entity may appeal to the Appellate Committee for redressal of his/their grievances with respect to the height permissible under these rules.

(3)
The cases for reference to the Appellate Committee specified in sub-rule (2) shall be received and processed by the corporate office at the headquarters of the Airports Authority in New Delhi.

12.
Responsibilities of local authorities and airport operators.

(1)
For the effective verification, monitoring and controlling the obstructions around the airports, it shall be the responsibility of the Local, Municipal or Town Planning and Development authorities and the airport operator to ensure that the height of the structures and their locations are in accordance with the approved building plans and the No Objection Certificate issued by the concerned designated officer or the authorised officer.

(2)
For the purposes of sub-rule (1), the Local, Municipal or Town Planning and Development authorities and the airport operator shall develop appropriate mechanism with necessary trained manpower and equipment so as to verify the height of the structures, site elevations and site location or coordinates in World Geodetic System 1984 (WGS84).

13.
Procedure to be followed in case of State owned and private airports not licensed by Directorate General of Civil Aviation.

(1)
In case of State owned or private aerodromes not licensed by the Directorate General of Civil Aviation, the concerned State Government shall be responsible for the protection of obstacle limitation surfaces at such airports:
Provided that the designated officer shall give guidance to the State Government on the protection of obstacle limitation surfaces, whenever such guidance is sought by the concerned State Government.

14.
Development and up gradation of aerodromes.

(1)
The approved master plan of the aerodromes shall be considered for drawing and protecting the various obstacle limitation surfaces to ensure its development and future expansion or up-gradation.

(2)
The designated officer or the authorised officer, before issuing the No Objection Certificate in respect of development or upgradation of any aerodrome (including its runway dimension), shall take into consideration the proposed communication, navigation and surveillance (CNS) facilities and the procedure for Air Navigation Service Operations (PANS-OPS) for height clearance at a given airport.

(3)
Necessary consultation with the concerned stakeholders shall be carried out by the airport developer, airport operator or by the Air Navigation Service provider, as the case may be, at the time of development of master plan of a Greenfield airport or planning of major airport expansion or the installation of new communication, navigation and surveillance facilities at the existing airports.

(4)

The aerodrome developer or operator and ANS provider, as the case may be, shall submit the approved master plan of the aerodrome and the proposed development or up gradation of any aerodrome (including its runway dimensions, communication, navigation and surveillance (CNS) facilities and the procedure for Air Navigation Service Operations (PANS-OPS) to the concerned designated officer).

15.

Procedure in case of violations.

- The cases of violations where the height of any existing building, structure or tree on any land within the limits specified in rule 4 exceeds the height specified in Schedule I and Schedule II, or any other violation arising out of non-compliance of the provisions of these rules, shall be dealt in accordance with the provisions of the Aircraft (Demolition of Obstructions caused by Buildings and Trees etc.) Rules, 1994.

16.

Savings.

- Nothing in these rules shall affect the height clearances assessed and duly issued under the notifications issued by the Government of India in the Ministry of Civil Aviation vide notification numbers S.O. 84(E) dated the 14th January, 2010, and S.O 1589(E) dated the 30th June, 2008, during their assessment validity period of eight years for the buildings and twelve years for the structures such as masts, chimney and towers etc., within which the applicants have to complete the structures and obtain the completion certificate from the concerned authorities:

Provided that in cases where the construction work has not started during the initial validity period of five years for the buildings or within seven years for the structures such as mast, chimney, etc., revalidation shall not be considered and the height of such buildings or structures shall be reassessed in accordance with the provisions of these rules.

I

Purpose: Schedule - 1 indicates the No Construction Zones (NCZ) i.e. the areas around the Aeronautical Ground Aids (AGA) and Communication Navigation & Surveillance (CNS) facilities which need to be kept free from all obstructions for the safety and regularity of aircraft operations.

1. Runway

1.1

Runway Strip: The Land area specified below shall be completely free from all obstacles as provided hereunder (Refer Appendix-1 of Schedule - I):-

1.1.1

. The land comprising within the Runway strip of uniform width of 150 meters on either side of centerline which extends to 60 meters beyond each extremity of Runway, along the extended centerline of a Runway of code 3 or code 4, equipped with Instrument Approach Procedure.

1.1.2

. The land comprising within the Runway strip of uniform width of 75 meters on either side of centerline which extends to 60 meters beyond each extremity of Runway, along extended centerline of the Runway of code 1 or 2, equipped with Instrument Approach Procedure and for non-Instrument runway of code 3 or 4.

1.1.3

. The land comprising within the Runway strip of uniform width of 40 meters on either side of centerline which extends to 60 meters beyond each extremity of Runway, along extended centerline of the non-Instrument Runway of code 2.

1.1.4

. The land comprising within the Runway strip of uniform width of 30 meters on either side of centerline which extends to 30 meters beyond each extremity of Runway, along extended centerline of the non-instrument runway of code 1.

1.2

Installation of Extra High Tension, High Tension lines shall not be permitted within 1500 metres of the Inner edge of the approach and take-off climb surface.

2. Frangibility Requirement:

2.1

Any equipment or installation required for air navigation purposes which must be located:

(a)

On that portion of the runway strip within:

i) 75 meters of the Runway centerline where the Runway code is 3 or 4 or

ii) 45 meters of the Runway centerline where Runway code is 1 or 2; or

(b)

on a runway end safety area, a taxiway strip or within the distances specified in Civil Aviation Requirements Section-4, Series 'B', Part I Aerodrome Design and Operations or

(c)

on a clearway and which would endanger an aircraft in the air, shall be frangible and mounted as low as possible.

2.2

Any equipment or installation required for air navigation purposes which must be located on or near a strip of precision approach Runway ILS category I, II or III and which-

(a)

is situated on that portion of the runway strip within 77.5 meters of the Runway centerline where the code number is 4 and code letter is F; or

(b)

is situated within 240 meters from the end of the runway strip and within-

(i)

60 meters of the extended runway centerline where Runway code is 3 or 4

(ii)

45 meters of the extended Runway centerline where Runway code is 1 or 2; or

(iii)

penetrates the inner approach surface, the inner transitional surface or the balked landing surface, shall be frangible and mounted as low as possible.

3. Communication, Navigation and Surveillance (CNS) Facilities

3.1

Very High Frequency Omni Range (VOR)/collocated Distance measuring Equipment (DME) and Very High Frequency Direction Finder (VHF DF): A land area within the 300 meters radius of the facility.

3.2

Localizer or LLZ (a component of ILS, providing azimuth guidance): the land area bounded by the following namely (Refer diagram at Appendix-L of Schedule VIII) :-

3.2.1

. A line 300 meters in the direction of approach or nearest end of the runway, whichever is greater from localizer antenna and perpendicular to the runway.

3.2.2

. A line 60 meters from the centerline of localizer antenna on both side and parallel to the runway.

3.2.3

. A line containing centre of localizer antennas and perpendicular to the runway; and

3.2.4

. Area within circle of 75 meters radius with centre at middle of the antenna system;

3.3

Glide Path (a component of ILS providing vertical guidance): the area bounded by the following, namely (Refer diagram at Appendix-K of Schedule VIII):-

3.3.1

. A line 300 meters in the direction of approach from the glide path facility;

3.3.2

. A line containing glide path antenna and perpendicular of runway;

3.3.3

. Near edge of the runway from the glide path;

3.3.4

. A line 30 meters in the directions away from the runway and parallel to it.

3.4

Locators or Markers Beacons: The land within a radius of 30 meters of the site of markers and locator beacons.

3.5

Airport Surveillance Radar (ASR): No structure will be permitted on the land above the level of 3 meters below the pedestal height up to the distance of 500 meters from Radar antenna.

3.6

Air Routes Surveillance Radar (ARSR): No structure will be permitted on the land above the level of 5 meters below the pedestal height up to the distance of 200 meters from Radar antenna.

3.7

Monopulse Secondary Surveillance Radar/ Secondary Surveillance Radar (MSSR/SSR): The distance and the height restriction shall be the same as in respect of the Airport Surveillance Radar or Air Routes Surveillance Radar, depending upon operational usage.

3.8

Microwave Link: On corridor of 30 meters on either side of the direct line of azimuth and 10 meters below from the direct line of sight in the vertical plane;

3.9

Ultra High Frequency (UHF) Link: On a corridor of 30 meters on either side of the direct line of the azimuth and 10 meters below from the direct line of sight in the vertical plane.

3.10

En-route Beacons: Land within a radius of 30 meters around the antenna.

3.11

Remote Receiver: Land within a radius of 1525 meters of the site.

3.12

Stand-alone Distance Measuring Equipment / Automatic Dependence Surveillance - Broadcast (DME/ADS-B): No structure will be permitted on land above the level of 3 meters below the antenna base upto a distance of 150 meters from the antenna.

3.13

Airport Surface Detection Equipment (ASDE) or Surface Movement Radar (SMR): No structure will be permitted on the land above the level of 2 meters below the antenna base up to the distance of 200 meters from Radar antenna.

3.14

Advanced Surface Movement Guidance and Control Systems (A-SMGCS) Reference Transmitter: No structure will be permitted on the land above the level of 5 meters below the antenna base up to the distance of 200 meters from Radar antenna.

3.15

A-SMGCS Multi-Lateration (MLAT): No structure will be permitted on the land above the level of 2 meters below the antenna base up to the distance of 200 meters from Radar antenna.

3.16

Ground Based Augmentation System (GBAS) Reference Receiver: No structure will be permitted on land up to the distance of 100 meters from antenna.

3.17

Ground Based Augmentation System VHF Data Broadcast (GBAS VDB) station: No structure will be permitted on the land up to the distance of 300 meters from antenna

3.18

GBAS VDB monitoring station: No structure will be permitted on the land up to the distance of 300 meters from antenna.

3.19

Global Position System (GPS) Pseudolite Restriction: No GPS Pseudolite shall be used within the approach funnel of any runway or within the airport where GNSS/GBAS based operation has been planned/exist.

3.20

Global Navigation Satellite System (GNSS) repeater restriction: No GNSS repeater shall be installed/use in approach funnel and within the 500 meter from the basic strip where GNSS/GBAS based operation has been planned/exist.

4. Definitions and Explanation. - Some of the definitions of the terms used in the notifications have been provided below. For other terms, the Civil Aviation Requirements Section-4, Series 'B', Part I Aerodrome Design and Operations, ICAO annex 14, Annex 10 and Doc.8168 may be referred.

i) Runway: A defined rectangular area on a land aerodrome prepared for the landing and take-off of aircraft.

ii) Runway end safety area (RESA). An area symmetrical about the extended runway centre line and adjacent to the end of the strip primarily intended to reduce the risk of damage to an aeroplane undershooting or overrunning the runway.

iii) "Runway Strip" A defined area including the runway and stopway, if provided, intended:

a) to reduce the risk of damage to aircraft running off a runway; and

b) to protect aircraft flying over it during take-off or landing operations.

iv) "Runway Code", means the Runway Code number specified in column (1), in relation to the Runway length specified in column (2), of the Table below:-

Table 1.1 Dimension Of Runway Strip

Runway

Instrument Runway

Non-Instrument Runway

Runway Code

Aerodrome Reference Field Length (ARFL) in (Meter)

Width Extending laterally on either side of Runway Centre Line
(Meter)

Length beyond Runway End/ Stop way (Meter)

Width Extending laterally on either side of Runway Centre Line
(Meter)

Length beyond Runway End/ Stop way (Meter)

(1)

(2)

(3)

(4)

(5)

(6)

1.

<800

75

60

30

30

2.

800<1200

75

60

40

60

3.

1200<1800

150

60

75

60

4.

1800 & above

150

60

75

60

v) "Approach funnel" in relation to (Refer Appendix-1 of Schedule-I):-

(a)

Instrument Runway Code 3 and 4, means the area in the shape of an isosceles trapezium having the longer parallel side 4800 meters long (2400 meters on either side of the extended centerline of the runway) and smaller parallel side 300 meters long (150 meters on either side of the extended centerline of the runway), where the smaller and longer parallel sides are placed at a distance of 60 meters and 15060 meters respectively, from the end of the runway and at right angles to the extended centerline;

(b)

Instrument Runway (Precision) Code 1 and 2, means the area in the shape of an isosceles trapezium having the longer parallel side 4650 meters long (2325 meters on either side of the extended centerline of the runway) and smaller parallel side 150 meters long (75 meters on either side of extended centreline of the runway) where the smaller and longer parallel sides are placed at a distance of 60 meters and 15060 meters respectively, from the end of the runway and at right angles to the extended centerline;

(c)

Instrument Runway (Non Precision) Code 1 and 2, means the area in the shape of an isosceles trapezium having the longer parallel side 900 meters long (450 meters on either side of the extended centerline of the runway) and smaller parallel side 150 meters long (75 meters on either side of the extended centerline of the runway), where the smaller and longer parallel sides are placed at a distance of 60 meters and 2560 meters respectively, from the end of the runway and at right angles to the extended centerline;

(d)

Non-Instrument Runway Code 3 and 4, means the area in the shape of an isosceles trapezium having the longer parallel side 750 meters long (375 meters on either side of the extended centerline of the runway) and smaller parallel side 150 meters long (75 meters on either side of the extended centerline of the runway), where the smaller and longer parallel sides are placed at a distance of 60 meters and 3060 meters respectively, from the end of the runway and at right angles to the extended centerline;

(e)

Non-Instrument Runway Code 2, means the area in the shape of an isosceles trapezium having the longer parallel side 580 meters long (290 meters on either side of the extended centerline of the runway) and smaller parallel side 80 meters (40 meters on either side of extended centreline of the runway) where the smaller and longer parallel sides are placed at a distance of 60 meters and 2560 meters respectively, from the end of the runway and at right angles to the extended centerline;

(f)

Non-Instrument Runway Code 1 means the area in the shape of an isosceles trapezium having longer parallel side of 380 meters long (190 meters on either side of the extended centreline of the runway) and smaller parallel side 60 meters (30 meters on either side of extended centreline of the runway) where the smaller and longer parallel sides are placed at a distance of 30 meters and 1630 meters respectively from the end of the runway and at right angles to the extended centreline. The diagrams of runway strip and approach funnel of instrument runway code 1,2,3 and 4 and non-instrument runway code 3 and 4 have been shown in the Appendix-1 of Schedule-I ;

vi) "Instrument Runway" means a runway served by visual aids and non-visual aids providing directional guidance adequate for a straight in approach and intended for the operation of aircraft using instrument approach procedures;

vii) Non-Precision Approach Runway - means an instrument runway served by visual aids and a nonvisual aid providing at least directional guidance adequate for a straight-in approach;

viii) Precision approach runway, category I - An instrument runway served by Instrument Landing System and/or MLS and visual aids intended for operations with a decision height not lower than 60 meters and either a visibility not less than 800 meters or a runway visual range not less than 550 meters.

ix) Precision Approach Runway, category II - An instrument runway served by Instrument Landing System and or MLS and visual aids intended for operations with a decision height not lower than 60 meters but not lower than 30 meters and a runway visual range not less than 350 meters.

x) Precision Approach Runway, Category III - An instrument runway served by Instrument Landing System (ILS) and/or MLS to and along with surface of the runway and -

(a)

ILS CAT IIIA - intended for operations with a decision height lower than 30 meters, or no decision height and a runway visual range not less than 200 meters.

(b)

ILS CAT IIIB - intended for operations with a decision height lower than 15 meters, or no decision height and a runway visual range less than 200 meters but not less than 50 meters.

(c)

ILS CAT IIIC - intended for operations with no decision height and no runway visual range limitations.

xi) "Non-Instrument Runway" means a runway intended for operations of the aircraft using visual approach procedure.

xii) "Very High Frequency Omni Range, Terminal Very High Frequency Omni Range, And Doppler Very High Frequency Omni Range" means the facilities operating in the Very High Frequency band of frequencies 112 to 118 MHz, radiate signals whereby an aircraft with the help of an instrument in its cockpit when tuned to the ground equipment frequency automatically gets its direction with respect to the facility and helps an aircraft to navigate on a predetermined course or home to an airport served by the facility.

xiii) "Instrument Landing System (ILS)" means the facility which serves to help an aircraft to make a safe landing on the runway in conditions of poor visibility and comprises of the following component facilities, namely:-

(a)

Localizer facility which radiates Very High Frequency Signals which when picked up by an aircraft guide it onto the centerline of the runway in the horizontal plane and is normally situated about 305 meters from the runway end;

(b)

Glide Path facility radiates Ultra High Frequency signals and is normally situated about 275 meters to 305 meters from the runway threshold and offset about 122 meters to 137 meters from the centerline of the runway and provides the glide angle information to a landing aircraft with the help of an instrument in the cockpit which when tuned to the glide path frequency indicates whether the aircraft is flying up or down or along the correct glide angle;

(c)

Outer Marker or Outer Locator facility operating on 75 MHz in the Very High Frequency band is normally installed along the extended centerline of the runway at a distance between 3.5 and 6 nautical miles (1 nautical mile=1853 meters) and produces radiation pattern to indicate the landing aircraft, the pre-determined distance from the threshold along the Instrument Landing System glide path;

xiv) "Radar" includes-

(a)

Airport Surveillance Radar (ASR) which is a radar facility serving an aerodrome to scan the air traffic within 50 to 60 nautical miles of the aerodrome;

(b)

Air Routes Surveillance Radar (ARSR) or Secondary Surveillance Radar is a high power long-range radar covering a distance of 200 nautical miles approximately and it scans air traffic to a larger distance than Airport Surveillance Radar;

xv) Communication and Navigational facilities include-

(a)

Microwave Link which is a radio facility whereby mostly intelligence/data is carried to the Air Traffic Control Display site;

(b)

Ultra High Frequency Link which is a radio relay facility operating in Ultra High Frequency Band;

(c)

Beacons which are radio transmitters operating in the Medium Frequency band from 200 to 400 KHz radiating omni directionally in the horizontal plane and an aircraft equipped with a suitable cockpit instrument can get its location automatically with respect to this facility.

(d)

Remote Receivers which are radio receiving stations (HF Band) installed at remote site away from factory or industrial areas to avoid interference link man-made static, etc.

Note: Location of Navigational Aids shall be determined as per the provisions of Annex-10 of International Civil Aviation Organization.

Appendix-1 to Schedule-I

II

Purpose:

The height or permissible elevation for the structure, requiring grant of NOC, shall be calculated based upon the

International Civil Aviation Organization (ICAO) Annex 14 Obstacle Restriction and Removal, Annex 10 the Radio Communication, Navigation and Surveillance (CNS) aids and Doc 8168, Vol II defining the operational requirements for minimum altitudes of various segments of published or proposed instrument approach procedures.

This annexure-II defines various OLS surfaces, requirements w.r.t. CNS and PAN-OPS, procedure to be followed while applying and processing the NOC for height clearance.

1. Obstacle Limitation Surfaces (based on ICAO Annex 14 and DGCA India Civil Aviation Requirements (CARs) on Aerodrome Design and Operations) are as under:

1.1

Take-off climb surface - The dimensions of the take-off climb surface shall not be less than the dimensions specified in the table given below except that if a runway is meant for takeoff, a lesser length may be adopted for the takeoff climb surface where such lesser length would be consistent with procedural measures adopted to govern the outward flight of aeroplanes.

Table 2.1 - Dimensions and Slopes of Obstacle Limitation Surfaces

(Runways Meant for Take-Off)

Surface and dimension*

Code Number

(1)

1

2

3 or 4

(2)

(3)

(4)

Take Off Climb

Length of inner edge

60 meters

80 meters

180 meters

Distance from runway end

30 meters

60 meters

60 meters

Divergence (each side)

10%

10%

12.5%

Final width

380 meters

580 meters

1200 meters

1800 meters**

Length

1600 meters

2500 meters

15000 meters

Slope

5%

4%

2%

* All dimensions are measured

horizontally

**When the intended track includes changes of heading greater than 15 degree for operations conducted in IMC, VMC by night

1.2

Transitional Surface

1.2.1

. The outer limit of the transitional surface is determined by its intersection with the plane containing inner horizontal surface and the slopes of transitional surfaces are as given below, namely:-

(i)

Precision Approach Runway

14.3% (1:7)

(ii)

Non Precision Runway

14.3% (1:7) for code 3 & 4

20% (1:5) for code 1 & 2

(iii)

Non-Instrument Runway

14.3% (1:7) for code 3 & 4

20% (1:5) for code 1 & 2;

The slope of the transitional surface shall be measured in a vertical plane at right angles to the centre line of the runway;

1.2.2

The elevation of a point on a lower edge shall be -

(a)

along the side of approach surface, equal to the elevation of approach surface at that point; and

(b)

along the strip, equal to the elevation of nearest point on the centre line of the Runway or its extension;

1.3

Approach Surface

1.3.1

The approach surface shall be established for each runway strip in the direction of intended landing of the aeroplanes and the limits and slopes are given table below:

1.3.1.1

Instrument Runway

Inner Edge of Approach Surface:

Length of Inner edge

- 150 meters for Code No. 1 and 2

- 300 meters for Code No. 3 and 4

Distance from runway threshold

- 60 meters

Divergence

- 15% on either side

Length & Slope of Approach Surface: as given in table 2.2

Table 2.2 - Approach Surface Slope of Instrument Runway

Runway

Precision Approach Runway

Non Precision Approach Runway

Code No.

Aerodrome Reference Field Length (meter)

First Section Length & (Meter) Slope

Second Section Length (Meter) & Slope

First Section Length (Meter) & Slope

Second Section Length (Meter)& Slope

Horizontal Section (Meter)

1.
<800
3000
2.5%
12000**
3%
2500
3.33%
-
-
-
-
2.
800<1200
3000
2.5%
12000**
3.00%
2500
3.33%
-
-
-
-
3.
1200<1800
3000
2%
3600
2.5%
3000
2%
3600
2.5%
8400*
4.
1800 and Above
3000
2%
3600
2.5%
3000
2%
3600
2.5%
8400*

* Total length of approach surface for runway code number 3 and 4 with precision and non-precision shall be 15000 meters.

** Total length of approach surface for Precision approach Runway Code number 1 and 2 shall be 15000 meters.

1.3.1.2

Non-Instrument runway

Inner Edge of Approach Surface:

Length of Inner edge

- 60 meters for Code No. 1

- 80 meters for code No. 2

- 150 meters for Code No. 3 and 4

Distance from runway threshold

- 30 meters for code 1

- 60 meters for code No. 2, 3 and 4

Divergence

- 10% on either side

Length & Slope of Approach Surface : as given in table 2.3

Table 2.3 - Approach Surface Slope of Non-Instrument Runway

Runway

Length and slope of approach surface

Code No.

Aerodrome Reference Field Length (meter)

Length (Meter)

Section Slope

1.

<800

1600

5%

2.

800<1200

2500

4%

3.

1200<1800

3000

3.33%

4.

1800 & above

3000

2.5%

1.3.1.3

Aerodrome where there are more than one runway with over-lapping approach areas and associated surface, most stringent of the two would be the applicable criteria.

1.3.1.4

For determining the approach, the physical extremities of the runway shall only be considered. In case of displaced threshold the permissible height shall be calculated based on approach surface and transitional surface with respect to the runway extremity or displaced threshold whichever is more restrictive.

1.3.1.5

At Aerodromes, where the proposals for runway extension exist, the requisite surface shall be determined from the proposed extension as well as from the existing runway strip/associated clearway, as applicable and the lower of the two elevations shall be permitted. The elevation of the associated runway extremity/displaced threshold/proposed extension of runway shall be the datum for approach surface.

1.3.1.6

The slope of the approach surface shall be measured in a vertical plane containing the centerline of the runway.

1.4

Inner Horizontal Surface (IHS)

1.4.1

Dimensions and permissible heights of Inner Horizontal Surface are given in the table below:

Table 2.4 - Dimensions and Permissible Heights of Inner Horizontal Surface

Runway

Instrument

Non-Instrument

Code No.

Aerodrome Reference Field Length (meter)

Radius (Meter)

Height (Meter)

Radius (Meter)

Height above Aerodrome Elevation (Meter)

1.

<800

3500*

45

2000*

45

2.

800<1200

3500*

45

2000*

45

3.

1200<1800

4000**

45

4000**

45

4.

1800 and Above

4000**

45

4000**

45

*For runway code number 1 and 2, radius of IHS shall be measured from the Aerodrome Reference Point (ARP).

** For runway code number 3 and 4, radius of IHS shall be measured from the runway extremity.

1.4.1.1

The reference datum for Inner-Horizontal Surface shall be the aerodrome elevation as defined in clause (g) of the Explanation to this notification.

1.4.1.2

For Runway code 3 and 4, the Inner Horizontal Surface shall be a composite pattern, which consists of two circular areas centered at the two ends with a radius of 4000 meters. These areas shall be joined tangentially to form an elliptical shape as shown in Appendix-A of Schedule VIII.

1.4.1.3

Where it is required to protect two or more widely spaced long runways, a more complex pattern involving four or more circular areas are formed. These areas should be joined tangentially by straight lines and the Inner Horizontal Surface shall be defined by the external limits of the resulting pattern (Refer Appendix-A of Schedule VIII).

1.4.1.4

When two aerodromes are close to each other with overlapping circuits the Inner Horizontal Surface will be drawn as

prescribed in para 1.4.1.3. The inner horizontal surface of these two aerodromes shall be joined tangentially to form one common Inner Horizontal Surface.

1.4.1.5

In case of common horizontal surface serving two aerodromes, the elevation of the Inner Horizontal Surface will be referenced to the lower of the two aerodromes.

1.5

Conical Surface

1.5.1

The conical surface shall be projected upwards and outwards from the periphery of the Inner Horizontal Surface. The slope 5% (1:20) of the conical surface shall be measured in a vertical plane perpendicular to the Inner Horizontal Surface. The reference datum for Conical Surface shall be the aerodrome elevations (Refer to Appendix-B of Schedule VIII for illustration of the various surfaces including the conical surface).

Note: Where a part of Inner Horizontal Surface and conical surface lies below the approach/take-off climb surface, the permissible heights shall be the lowest of the applicable surfaces.

1.6

Outer Horizontal Surface (OHS)

1.6.1

The Outer Horizontal Surface shall extend to 15000 meters from the Aerodrome Reference Point for Aerodrome with runway code 3 and 4.

1.6.2

In case of Aerodrome with runway Code 2, the Outer Horizontal Surface shall extend to 14740 meters from Aerodrome Reference Point for Instrument runways and 13740 meters for Non-Instrument runways.

1.6.3

Where combined Outer Horizontal Surface is established for two Aerodromes, the Outer Horizontal Surface shall be centered on the Aerodrome Reference Point of the Aerodrome of higher category.

1.6.4

Outer Horizontal Surface for Aerodrome with runway code No.1 shall not be established.

1.6.5

The Outer Horizontal Surface, would be defined such that the Conical Surface may continue to be extended at 5% slope to a point wherein the permissible maximum height of *300 meters (above aerodrome elevation) is reached and thereafter this surface is maintained upto 15 kilometers from Aerodrome Reference Point. Construction(s) protruding above these surfaces shall normally not be permitted. Obstructions existing in the area should be marked or lighted.

*Note: In case of Defence Aerodromes, the permissible maximum height in conical and OHS shall be 150 meters above aerodrome elevation.

1.6.6

In order to avoid abrupt vertical changes in surfaces, the surfaces beyond the conical surfaces will slope laterally at 1:7 from edges of the approach and take off surfaces between the permissible heights of 150 meters to 300 meters (For illustration refer to Appendix-B of Schedule VIII).

1.6.7

The datum for Outer Horizontal Surface shall be the aerodrome elevation.

1.7

The Inner Approach, Inner Transitional and Balked Landing Surfaces (collectively referred as Obstacle Free Zone or OFZ):

1.7.1

Obstacles Free Zone shall be established for a runway equipped with precision approach (ILS) category I, II and III operations. The zone shall be kept free from fixed objects other air navigation aids, which must be near the runway, to perform their function, mounted on light weight frangible fixtures.

Note: Obstacles Free Zone for runway code No. 1 and 2 are not established.

The dimensions and slopes of the Obstacles free zone (Code 3 and 4) are given below.

1.7.1.1

The inner approach surface

Width

120 meters

Distance from Threshold

60 meters

Length

900 meters

Slope

2%

1.7.1.2

. The inner transitional surface

Slope

33.3%

1.7.1.3

. Balked Landing Surface

Length of the Inner edge

120 meters

Distance from Threshold

1800 meters

Divergence

10%

Slope

3.33 %

2. Protection of Service volume of various Communication, Navigation and Surveillance Facilities (based on ICAO Annex 10 Navigational Aids)

2.1

Very High Frequency Omni Range (VOR), Terminal Very High Frequency Omni Range (TVOR), and collocated Very High Frequency Omni Range Distance Measuring Equipment (VOR DME) - No structure (located beyond the area of 300M radius as specified in Annexure I) shall subtend a vertical angle greater than 1.5 degree at the centre of the Very High Frequency Omni Range counterpoise from the horizontal plane passing through the counterpoise.

2.2

Stand-alone Distance Measuring Equipment (DME) - No steel towers, power lines, metal buildings (located beyond the area of 150M radius as specified in Annexure I) shall protrude elevation angle of 3 degree measured from the base of Distance Measuring Equipment antenna.

2.3

Localizer

2.3.1

Within ± 10 degrees azimuth in front of LLZ antenna, an object (located beyond the area specified in Annexure I) should not subtend an angle of elevation more than 0.75 degrees at the centre of antenna array.

2.3.2

Within ± 10 degrees to ± 35 degrees LLZ azimuth in front of antenna an object (located beyond the area specified in Annexure I) should not subtend an angle of elevation more than 1.1 degree.

2.3.3

Notwithstanding any thing in para 2.3.1 and 2.3.2, in all airports having/intended to have cat II and cat III ILS, all object in sector of ± 18 degree for medium aperture antenna localizer and $+ 15$ degree for wide aperture LLZ antenna, upto the distance of 1050M beyond threshold, to be analysed for their potential multipath effects on the performance of ILS.

2.4

Glide Path - Beyond areas specified in Annexure I and within ± 8 degrees azimuth in front of the glide path antenna (in the direction of approach), a building/structure should not subtend an angle of elevation of more than 1.1 degree at antenna base.

2.5

Airport Surveillance Radar (ASR);

2.5.1

Wherever airport is served or proposed to be served by a single ASR, following criteria shall be applicable:

2.5.1.1

Beyond 500 meters from particular Radar site, the height of the permissible structures may be increased at the rate of 0.05 per meter, upto a point wherein the height of the permissible structure does not protrude above the line drawn from a point 10% below the minimum sector altitude at the farthest point (from Radar site) or any other designated MSA at different distance in same sector whichever is closer to horizon, to the centre of antenna pedestal, considering the Minimum Sector Altitude (MSA) in that particular sector. Beyond the above stated point no large object would be permitted to protrude above the line drawn from a point 10% below the minimum sector altitude at the farthest point (from Radar site) or any other designated MSA at different distance in same sector whichever is closer to horizon to the centre of antenna pedestal depending on the minimum Sector Altitude in that particular sector (For illustration refer to Appendix-C of Schedule VIII).

Note: Large object means the structure/s in isolation or collectively subtending azimuth angle of 0.4 degree or above at Radar antenna. In case of cluster of buildings wherein the gap between the two adjacent buildings subtends an azimuth angle of less than 0.4 degree on the antenna pedestal, the entire cluster should be considered as one object.

2.5.2

Wherever airport is served or proposed to be served by Multiple Radars (more than one ASR), operational and integrated, following criteria shall be applicable:

2.5.2.1

In case only one ASR is installed and the proposed ASRs are yet to be operationalized and integrated, the existing ASR will be considered for height calculations as per the provisions of 2.5.1.

2.5.2.2

After multi radar system is operationalized and integrated, the maximum height permissible in the integrated system will be considered for calculation of height to the applicant. However, from the radar performance requirement point of view, the structures are to be examined, as follows, to ensure that there is no degradation of radar performance.

I. Within one kilometer of any ASR in the system, structures shall be examined from the respective radar as per para 2.5.1.

II. Between one and two kilometer, the metallic and large structures shall be examined from respective ASR as per para 2.5.1.

III. Structures which are Non-metallic and are not termed as large objects may be permitted to higher height as per IV below, subject to condition that other structure(s) in vicinity do not form cluster with the structure under examination.

IV. Objects beyond two kilometer from any one of the ASRs, highest permissible height among integrated & operational ASR sites shall be permitted as per para 2.5.1

Note: Above criterion will not be applicable for wind farms, high tension lines and electromagnetic source of interference.

2.6

Air Route Surveillance Radar (ARSR) - Beyond 200 meters from particular Radar site the height of the permissible structures may be increased at the rate of 0.05 meter per meter, upto a point wherein the height of the permissible structure does not protrude above an angle of elevation of more than 0.5 degree at the antenna pedestal or an angle equal to antenna tilt angle set during last flight inspection whichever is higher. Beyond the above stated point no large object would be permitted to protrude above the line drawn at an angle of 0.5 degree from antenna pedestal or an angle equal to antenna tilt angle set during last flight inspection whichever is higher. Large object means the structure subtending azimuth angle of 0.4 degree or above at Radar antenna. In case of cluster of buildings wherein the gap between the two adjacent buildings subtends an azimuth angle of less than 0.4 degree on the antenna pedestal, the entire cluster should be considered as one object (For illustration refer to Appendix-D of Schedule VIII).

2.6.1

Monopulse Secondary Surveillance Radar / Secondary Surveillance Radar (MSSR/SSR) Same as Air Surveillance Radar/Air Route Surveillance Radar depending on operational usage.

2.7

Automatic Dependence Surveillance-Broadcast (ADS-B) - Beyond 150m from particular ADS-B site, the height of the permissible structures does not protrude above the line drawn from a point 10% below the minimum sector altitude at the farthest point (from ADS-B site) or any other designated MSA at different distance in same sector whichever is

closer to horizon, to the centre of the antenna pedestal of ADS-B.

2.8

Advance Surface Movement Guidance and Control System (A SMGCS): - No structure should be built on the relevant area of the airport surface which blocks the line of sight between any of the sensors of the Advance Surface Movement Guidance and Control System and the relevant operational area. In case there is an operational or safety/security requirement to add a structure on the airport surface which may obstruct the line of sight between Surface Movement Radar (SMR) antenna/ sensors, AAI would augment the system to meet the Advance Surface Movement Guidance and control system operational requirement.

2.8.1

Surface Movement Radar (SMR) - Beyond the distance of 200 M. from SMR antenna, no object should protrude the line of sight to nearest point of designated coverage volume of said SMR.

2.8.2

Beyond the distance of 200 M from A-SMGCS Ref TX no object should protrude the line of site to nearest point of designated coverage volume of said ref TX and corridor between ref TX and MLAT of 5 Mx5 M.

2.8.3

Beyond the distance of 200 M from A-SMGCS MLAT no object should protrude the line of designated coverage volume of said ref TX and a corridor between ref TX and MLAT of 5 Mx5 M.

2.9

Indian Land Uplink Station (INLUS)/Indian National Reference Station (INRES) of GPS Aided Geo Augmented Navigation (GAGAN) System - No structure will be permitted to protrude the above the plane inclined at elevation angle of 2 degree from the horizontal surface drawn at the level of antenna of Indian Land Uplink Station and Indian National Reference Station of GPS Aided Geo Augmented Navigation system which is a part of Global Navigation Satellite System (GNSS).

2.10

Very High Frequency (VHF)/ Remote Controlled Air to Ground communication (RCAG) - no structure shall be allowed to protrude above the lowest line of sight of coverage of designated service volume of facility without proper mitigation.

2.11

Wind Turbine Generators/ Wind Farms - No Wind Turbine Generator/s shall be installed upto a distance of 10 KM in line of sight of the Radar Antenna of all Static Air Defence Radars and upto 8 KM from VOR and Airport Surveillance Radar (ASR).

2.12

Electricity Power Transmission Lines

2.12.1

No High Tension (HT) or Low Tension (LT) line shall be permitted to pass through the sensitive area of Localizer and glide path.

2.12.2

All HT lines will not be permitted to the following area until and unless these are shielded by permanent structures:-

- Localizer, with in ± 18 degree, all HT lines will be permitted only up to an angle of elevation of 0.5 degree from the localizer. If these HT line are on the radial, these may be permitted to 0.75 degree elevation. In the sector between ± 18 degree to ± 35 degree line may be permitted up to the elevation angle of 0.75 degree.
- Glide Path, all HT lines will be permitted only up to an angle of elevation of 0.5 degree from the Glide Path. If the HT line is on the radial, it may be permitted to 0.75 degree elevation.
- VOR, HT lines shall be permitted below 0.5 degree at counterpoise and if these lines are on the radial, they may be permitted up to 1 degree.
- RADAR, power line above 11 KVA and up to 100 KVA may not be permitted up to 1 km and above 100 KVA up to 2 KM

3. Procedure for Air Navigation Services Operation (PANS-OPS) criteria (based on ICAO Document 8168, Volume II):

3.1

In order to achieve the lowest possible operating minima for aircraft operation, it is necessary to protect not only the Annex 14 OLS but also to safeguard the PANS-OPS [ICAO Document 8168] Surfaces. The limit of PANS-OPS surfaces extend up to 30NM from the facility i.e. VOR or NDB serving the aerodrome. Considerations need to be given to the

objects which penetrate the PANSOPS surfaces, regardless whether or not they penetrate Annex 14 OLS. Such obstacle may result in an operational penalty like higher Obstacle Clearance Altitude/Height (OCA/H) and introduction of longer approach segment. Therefore, while examining the cases for issue of NOC from the considerations of Annex 14 and Annex 10 criteria as provided in para 1 and 2 above, the operational criteria needs to be considered based on the provisions of Documents 8168, Vol. - II. It needs to be ensured that the minimum altitudes of the following segments, published or the proposed, are not infringed:

- i. Minimum Sector Altitude (MSA)
- ii. Minimum Holding Altitude (MHA)
- iii. Minimum Vectoring Altitude (MVA)
- iv. Minimum Altitude of Initial and Intermediate Segments
- v. OCA/H (Straight-in-and Circling) for all aircraft categories
- vi. STARs /SIDs procedure altitude.
- vii. Basic ILS Surface

3.2

Criterion specified in Doc 8168 Volume II (PANS-OPS) for designing instrument procedures shall not be used for creating new structures as PANS-OPS surfaces are not intended to replace Annex 14 OLS as planning surfaces for creating new structures.

3.3

For the obstacles located even outside the limits of Annex 14 OLS, it shall be ensured that PANSOPS surfaces of the published instrument approach procedures are not penetrated.

Note 1: Instrument approach procedures of all the civil aerodromes in India have been published in the AIP India under the section "Aerodrome". In the published procedures, the minimum altitudes of the various segments of instrument approach procedures have been specified.

Note 2: The minimum obstacle clearance criteria are applied as per the provisions of International Civil Aviation Organization (ICAO) Document 8168 Volume II. Normally for minimum sector altitudes (Applicable upto 30 NM from the facility on which procedure is designed), minimum vectoring altitudes, minimum holding altitudes and for the initial approach an obstacle clearance of 1000 feet is applied.

Note 3: Final approach areas of Very High Frequency Omni Radio Range (VOR)/Non Directional Beacon (NDB) have been illustrated in Appendix-E of Schedule VIII).

4. Shielding criteria - The principle of shielding is applicable w.r.t. Natural Terrain, already penetrating one of the obstacle limitation surfaces of an airport and it is not likely to be removed. The shielding criteria as explained below is applicable w.r.t. AGA and CNS surfaces.

4.1

The principle of shielding will not to be applied in:

- I. Transitional surface area,
- II. Approach surface areas, within 4000 meters of the inner edge of approach surface.
- III. Inner Horizontal Surface (IHS), within a distance of 2500 meters from the runway centre line. In case of multiple runways, area encompassed by 2500M from centerline of all runways.

4.2

The following criteria shall be followed for the purpose of applying shielding criteria for the proposed structure with respect to existing natural terrain.

4.2.1

Proposed (shielded) object located beyond a distance of 2500M from runway centerline:

(i)

Draw a line joining the centre point of the plot to the nearest runway end (runway code no. 3 & 4) or ARP (code 1 & 2) as the case may be. Shielding will be applicable w.r.t. applicable terrain within the area bounded by the two lines drawn parallel to the above line, at a distance of 600M on either side. A line, across the highest point of applicable reference (shielding) terrain, perpendicularly to the above parallel lines shall be drawn to delineate the areas for different type of shielding i.e. negative or equal to the horizontal plane passing through top of reference terrain (For the illustrations refer to Appendix-F and Appendix-G of Schedule VIII).

(ii)

If the proposed structure is lying between the aerodrome and the reference terrain, a negative shielding of 10% shall be applicable. The shielding benefit of a horizontal plane, equal to reference terrain height, shall be provided in the area located in the opposite side away from the aerodrome (For the illustrations refer to Appendix-F and Appendix-G of Schedule VIII).

4.3

Communication Navigation Surveillance (CNS) Parameters: - For CNS facilities, shielding benefit could be provided to the structures in cases wherein such structures (shielded) are in the shadow of the highest terrain of permanent nature. Shadow for this purpose is defined as an area falling below a line drawn from the top and both the extremities of the terrain of permanent nature, to the facility and extrapolation of the same plane behind from the said obstacle.

5. Conduct of Aeronautical Study and CNS Simulation Study

5.1

The Aeronautical Study, as referred to in the Civil Aviation Requirements Section-4, Series 'B', Part I on Aerodrome Design and Operations and ICAO Annex 14, may be conducted to determine that the existing object or the proposed new object would not adversely affect the safety or significantly affect the regularity of operations of aeroplanes in pursuance of the ICAO provisions as given below:

Note 1: New objects or extensions of existing objects should not be permitted above the conical surface and the inner horizontal surface except when, in the opinion of the appropriate authority, after aeronautical study it is determined that the object would not adversely affect the safety or significantly affect the regularity of operations of aeroplanes.

Note 2: Existing objects above an approach surface, a transitional surface, the conical surface and inner horizontal surface should as far as practicable be removed except when, in the opinion of the appropriate authority, after aeronautical study it is determined that the object would not adversely affect the safety or significantly affect the regularity of operations of aeroplanes.

5.1.1

The request for aeronautical study shall be considered by the Member (Air Navigation Services), Airports Authority of India, on case to case basis.

5.1.2

Aeronautical Study shall not be carried out in Approach and Transition surfaces.

5.1.3

Aeronautical Study, as per the established guidelines, shall be carried out by AAI, ICAO or any other agency, approved for the purpose by Ministry of Civil Aviation.

5.1.4

Based on the Aeronautical Study report, including a revised height clearance if necessary, shall be communicated to the applicant by AAI.

5.1.5

Guidelines are available at NOCAS at www.aai.aero.

5.2

Communication Navigation Surveillance (CNS) Simulation study: In case any structure is required to be made within aerodrome premises (airside and city side) by the Aerodrome Operator which creates obstruction from CNS point of view, a simulation study could be carried out to study the impact of this structure on the performance of the relevant facility and in case the study confirms that the impact would not hamper the operability of the facility, such structure could be permitted within the aerodrome premises.

6. Procedure for determining the maximum permissible heights: - The following steps shall be taken for calculating the maximum permissible heights for cases where there is a requirement of NOC from AAI or from Defence Authorities.

6.1

ICAO Annex 14 Obstacle Limitation Surfaces Criteria:

6.1.1

The site of the proposed buildings/installations shall be marked on the zoning map of the aerodrome, prepared by the aerodrome operator, where Annex 14 surfaces have been drawn or plotted on the map generated by NOCAS based on the site co-ordinate(s) in WGS 84 system.

6.1.2

If the site location is within the approach/take off surface, the permissible applicable heights in the approach/take off

climb surface, transitional surface, Inner Horizontal Surface/conical surface shall be calculated.

6.1.3

If the site is located outside the approach/take off climb surface, the height shall be determined as per the location applicable to the relevant surface (Transitional, Inner Horizontal Surface, Conical or Outer Horizontal Surface).

6.2

ICAO Annex 10 Communication, Navigation and Surveillance (CNS) Criteria:

6.2.1

Determine the distance of the proposed site from the each communication, navigational and surveillance facility separately and calculate the applicable heights based on the provisions as contained in para 2 of Annexure II.

6.3

The permissible height from the above two criteria shall be the lowest as of 6.1 and 6.2 above.

6.4

Procedure for Air Navigation Service Operations (PAN-OPS) Criteria:

6.4.1

After having determined the combined applicable elevation, based on the OLS criteria and CNS criteria, it shall further be ensured that the PANS-OPS surfaces are not infringed and the minimum altitudes of the published/proposed segments of instrument approach procedures are fully protected. This has also been referred to at para 3 of this annexure.

6.4.2

The lowest elevation determined as above, based on the OLS, CNS and PANS-OPS criteria, shall be the permissible top elevation of the proposed structure for which No Objection Certificate may be issued by the designated officer of AAI or the Defence Authorities.

6.5

No Objection Certificate Application System (NOCAS) for applying for height clearance w.r.t. Civil Airports:

6.5.1

AAI has introduced "No Objection Certificate Application System (NOCAS)" accessible at the AAI website www.aai.aero for online submission of NOC application for height clearance. NOCAS carries out calculations w.r.t. OLS and CNS criteria based on site coordinates and elevation provided by the applicant. It is mandatory for the applicant to provide surveyed site coordinates in WGS 84 system and site elevation from a Govt. entity or a Govt. approved agency. The applicants are first required to register themselves online and only thereafter, they can submit their applications for NOC. On registering in NOCAS, a NOCAS ID is generated which can be used for future reference including status check of the application. Guidelines for online submission of NOC application for height clearance are available at NOCAS at www.aai.aero.

6.5.2

A table of permissible heights w.r.t. Annex 14 OLS criteria at different distances from the runway (Code 3 or 4 Instrument runway) at an airport are given at Appendix-L of Schedule VIII.

6.6

The permissible heights given therein are only indicative w.r.t. OLS criteria only; detailed calculations w.r.t. CNS and PANS-OPS criteria are needed to arrive at the actual height permissible, which may be lower than the indicated.

7. Definitions and Explanation - Description of Annex 14 Obstacle Limitation Surface for the purpose of the Schedule II shall be as given hereunder and the illustrations in respect thereof are given in Appendix-H, Appendix-I and Appendix-J of Schedule VIII.

(a)

Conical Surface - A surface sloping upwards and outwards from the periphery of the inner horizontal surface.

The limits of the conical surface shall comprise:

(i)

a lower edge coincident with the periphery of the inner horizontal surface; and

(ii)

an upper edge located at a specified height above the inner horizontal surface.

The slope of the conical shall be measured in a vertical plane perpendicular to the periphery of the inner horizontal surface.

(b)

Inner Horizontal Surface - A surface located in a horizontal plane above an aerodrome and its environs. The radius of outer limits of the inner horizontal surface shall be measured from a reference point or points established for such purpose.

(c)

Inner Approach Surface - A rectangular portion of the approach surface immediately preceding the threshold. The limits of the inner approach surface shall comprise:

(i)

an inner edge coincident with the location of the inner edge of the approach surface but of its own specified length;

(ii)

two sides originating at the ends of the inner edge and extending parallel to the vertical plane containing the centerline of the runway; and

(iii)

an outer edge parallel to the inner edge.

(d)

Inner Transitional Surface - A surface similar to the transitional surface but closer to the runway.

The limits of an inner transitional surface shall comprise:

(i)

a lower edge beginning at the end of the inner approach surface and extending down the side of the inner approach surface to the inner edge of that surface, from there along the strip parallel to the runway centerline to the inner edge of the balked landing surface and from there up the side of the balked landing surface to the point where the side intersects the inner horizontal surface; and

(ii)

an upper edge located in the plane of the inner horizontal surface.

(e)

Balked Landing Surface - an inclined plane located at a specified distance after the threshold extending between the inner transitional surfaces. The limits of the balked landing surface shall comprise:

(i)

an inner edge horizontal and perpendicular to the centre line of the runway and located at a specified distance after the threshold;

(ii)

two sides originating at the ends of the inner edge and diverging uniformly at a specified rate from the vertical plane containing the centre line of the runway; and

(iii)

an outer edge parallel to the inner edge and located in the plane of the inner horizontal surface.

(f)

Take-Off Climb Surface (Annex 14) - The surface shall be established for a runway meant for take-off. The limits of the take-off climb surface shall comprise:

(i)

an inner edge horizontal and perpendicular to the centre line of the runway and located either at a specified distance beyond the end of the runway or at the end of the clear way when such is provided and its length exceeds the specified distance;

(ii)

two sides originating at the ends of the inner edge and diverging uniformly at a specified rate from the take-off to specified final width and continuing thereafter at that width for the remainder of the length of the take-off climb surface; and

(iii)

an outer edge horizontal and perpendicular to the specified take-off track.

(g)

Aerodrome Elevation- The elevation of the highest point of the landing area.

(h)

Aerodrome Reference Point - The designated geographical location of an Aerodrome.

(i)

Threshold - The beginning of that portion of the runway usable for landing.

(j)

Displaced Threshold - A threshold-not located at the extremity of a runway.

(k)

Frangible Object - An object of low mast designed to break, distort or yield on impact so as to present the minimum hazard to aircraft.

(l)

Obstacle - All fixed (whether temporary or permanent) and mobile objects, or parts thereof, that are located on an area intended for surface movement or aircraft or that extend above a defined surface (indicated in annexure IV) intended to protect aircraft in-flight.

(m)

Obstacle Free Zone (OFZ) - The airspace above the inner approach surface, inner transitional surfaces and balked landing surface and that portion of the strip bounded by these surfaces, which is not penetrated by any fixed obstacle other than low mast and frangible mounted one, required for air navigation purposes.

(n)

Runway - a defined rectangular area on a land aerodrome prepared for the landing and take off of the aircraft.

(o)

Runway End Safety Area (RESA) - An area symmetrical about the extended runway centre-line and adjacent to the end of the strip primarily intended to reduce the risk of damage to an aeroplane undershooting or overrunning the runway.

(p)

Runway Strip - A defined area including the runway and stop-way, if provided, intended:-

(i)

To reduce the risk of damage to aircraft running off a runway; and

(ii)

To protect aircraft flying over it during take off or landing operations.

(q)

Clearway - A defined rectangular area on the ground or water under the control of the appropriate authority selected or prepared as a suitable area over which an aeroplane may make a portion of its initial climb to specified height.

(r)

Stop-way - A defined rectangular area on the ground at the end of take off run available prepared as suitable area in which an aircraft can be stopped in case of an abandoned take-off.

(s)

Take-off Runway - a runway intended for take-off only.

(t)

Obstacle Clearance Altitude/Height (OCA/H) - The lowest altitude or the lowest height above the elevation of the relevant runway threshold or the aerodrome elevation as applicable used in establishing compliance with appropriate clearance criteria.

(u)

Declared Distances:-

(i)

Take Off Run Available (TORA) - The length of the runway declared available and suitable for the ground run of an aeroplane taking off.

(ii)

Take Off Distance Available (TODA) - The length of take-off run available plus the length of clearway, if provided.

(iii)

Accelerate Stop Distance Available (ASDA) - The length of take-off run available plus the length of stop-way, if provided.

(iv)

Landing Distance Available (LDA) - The length of the runway declared available and suitable for the ground run of an aeroplane landing.

(v)

Critical Area - Critical area is an area of defined dimensions about the localizer and glide path antenna where vehicles including aircrafts are excluded during Instrument Landing System (ILS) operations. The critical area is protected because the presence of vehicles and/or aircraft inside its boundary will cause unacceptable discrepancies to the Instrument Landing System (ILS) signal in space.

III

Aerodromes Operated By Airports Authority India And Joint Venture Companies

S. No.

Airport

State

Coordinates

Aerodrome Elevation in Meters

Runway

Dimension in Meters

Operator

(1)

(2)

(3)

(4)

(5)

(6)

(7)

1

Agartala

Tripura

235326N

0911421E

14.63

18/36

2286 X 45

AAI

2

Agatti

Lakshwadeep Islands (U.T.)

104926N

0721037E

4

04/22

1204 X 30

AAI

3

Ahmedabad (SVBPI Airport)

Gujarat

230416.28N

0723735.15E

57.44

05/23

3505 X 45

AAI

4

Aizawl (Turial)

Mizoram
234443N
0924822E
334
01/19
1190 X 27
AAI
5
Akola
Maharashtra
204152N
0770332E
305
10/28
1219 X 45
AAI
6
Amritsar (Raja Sansi)
Punjab
314217N
0744807E
231
16/34
3658 X 45
AAI
7
Asansol
West Bengal
2340N
08701E
98
10/28
1826 X 45
AAI
8
Aurangabad (Chikal Thana)
Maharashtra
195152N
0752351E
582
09/27
2835 X 45
AAI
9
Balurghat
West Bengal
251547N
0884754E
24
09/27
1097 X 30

AAI
10
Barapani Shillong
Meghalaya
254212N
0915841E
887
04/22
2286 X 45
AAI
11
Behala
West Bengal
223022N
0881748E
3
18/36
861 X 30
AAI
12
Belgaum (Sambra)
Karnataka
155131N
0743704E
759
08/26
1830 X 45
AAI
13
Bengaluru International Airport (BIAL) Devanhalli
Karnataka
131155.92N
0774219.70E
914.68
09/27
4000 X 45
BIAL
14
Bhavnagar
Gujarat
214515N
0721126E
13
07/25
1920 X 45
AAI
15
Bhopal (Raja Bhoj Airport)
Madhya Pradesh
231713N
0772013E

521.82
12/30
2744 X 45
AAI
16
Bhubneshwar (Biju Patnaik Airport
Orissa
201448N
0854907E
42.06
14/32
2743 X 45
AAI
17
Bilaspur
Chattisgarh
220000N
0820400E
274
06/24
17/35
1448 X 45
1455 X 45
AAI
18
Chakulia
Jharkhand
222736N
0864237E
129
17/35
2221 X 45
AAI
19
Chennai
Tamil Nadu
125941.7N
0801031.8E
15.85
07/25
12/30
3658 X 45
2890 X 45
AAI
20
Cochin International Airport Ltd (Cial)
Kerala
100914N
0762425E
9.14
09/27

3400 X 45
CIAL
21
Coimbatore (Peelamedu)
Tamil Nadu
110137N
0770230E
404
05/23
2990 X 45
AAI
22
Cooch Behar
West Bengal
261946.8N
089281.6E
42
04/22
1069 X 30
AAI
23
Cuddapah
Andra Pradesh
1431N
07847E
131
11/29
1098 X 18
AAI
24
Deesa (Palanpur)
Gujarat
241604N
0721218E
145
06/24
1008 X 30
AAI
25
Dehradun (Jollygrant)
Uttarakhand
301126N
0781056E
565
08/26
2140 X 45
AAI
26
Delhi IGI Airport (Palam)
Delhi
283407.42N

0770643.69E
236.83
10/28
09/27
11/29
3810 X 45
2813 X 45
4430 X 60
AAI Leased to DIAL
27
(Mohanbari)
Assam
272852N
0950105E
110
05/23
2310 X 45
AAI
28
Dimapur
Nagaland
255300N
0934616E
148.43
12/30
2290 X 45
AAI
29
Donakonda
Andhra Pradesh
1550N
7930E
142
04/23
915 X 30
AAI
30
Gondia
Maharashtra
2131N
08020E
311.16
05/23
2290 X 45
AAI
31
Guwahati (LGBI Airport)
Assam
260618N
0913508E
49.38

02/20
3103 X 45
AAI
32
Gaya
Bihar
244453N
0845633E
115.82
10/28
2286 X 45
AAI
33
Hubli
Karnataka
152147N
0750508E
661.72
08/26
1674 X 30
AAI
34
Hyderabad (Begumpet)
Andra Pradesh
172709N
0782750E
531
09/27
3103 X 45
AAI
35
Hyderabad International Airport(HIAL) Shamsabad
Andhra Pradesh
171426N
0782544E
617
09/27
4260 X 60
JV Airport operated by HIAL
36
Imphal (Tuliha)
Manipur
244551N
0935358E
774.2
04/22
2746 X 45
AAI
37
Indore Devi Ahilyabai Holkar Airport
Madhya Pradesh

224324N
0754820E
563.88
07/25
2750 X 45
AAI
38
Jabalpur
Madhya Pradesh
231100N
0800337E
495
06/24
1988 X 45
AAI
39
Jalgaon
Maharashtra
205741.74N
0753728.43E
256
09/27
1700 X 45
AAI
40
Jaipur (Sanganer)
Rajasthan
264927N
0754809E
385
09/27
3500 X 45
AAI
41
Jhansi
Uttar Pradesh
2529N
07834E
244
15/33
1295 X 45
AAI (Under Army)
42
Jharsuguda
Orissa
215451N
0840303E
228
06/24
1882 X 45
AAI

43
Jogbani
Bihar
2618N
8718E
59
09/27
1525 X 152
AAI
44
Juhu (Mumbai)*
Maharashtra
190548N
0725004E
2.74
08/26
16/34
1133 X 30
732 X 20
AAI
45
Kailashahar
Tripura
241828N
920033E
24
03/21
900 X 30
AAI
46
Kamalpur
Tripura
240754N
0914851E
39
01/19
1372 X 30
AAI
47
Keshod
Gujarat
211852N
701610E
51
05/23
1372 X 45
AAI
48
Kandla
Gujarat
230642N

0700605E
29
05/23
1524 X 30
AAI
49
Khandwa
Madhya Pradesh
21 51N
76 20E
329
10/28
975 X 46
AAI
50
Khowai
Tripura
240342N
913627E
29
18/36
915 X 30
AAI
51
Gaggal (Kangra)
Himachal Pradesh
320955N
0761543E
759.6
15/33
1372 X 30
AAI
52
Kanpur (Civil)
Uttar Pradesh
262625N
0802153E
125
10/28
1082 X 45
AAI
53
Khajuraho
Madhya Pradesh
244912N
0795506E
217.4
01/19
2286 X 45
AAI
54

Kishangarh

Rajasthan

2636N

07449E

440

05/23

1700 X 45

AAI

*Proposed construction in the approach path of runway 26 at Juhu shall be governed by the recommendation of the report of the Joglekar Committee.

*Sites lying in IHS of Juhu may be considered upto IHS of Santa Cruz in accordance with report of the study of Juhu IHS.

55

Kota

Rajasthan

250935N

0755056E

273

08/26

1243 X 38

AAI

56

Kozhikode (Calicut)

Kerela

110816N

0755702E

98.76

10/28

2860 X 45

AAI

57

Kolhapur

Maharashtra

163955N

0741729E

606.5

07/25

1370 X 45

AAI

58

Kolkata (NSCBI Airport)

West Bengal

2239114N

0882648E

7

01R/19L

01L/19R

3627 X 45

3270 X 45

AAI
59
Kullu-Manali (Bhuntar)
Himachal Pradesh
315237N
0770919E
1088.8
16/34
1052 X 30
AAI
60
Lalitpur
Uttar Pradesh
244258N
0782503E
367
10/28
1890 X 45
AAI
61
North Lakhimpur (Lilabari)
Assam
271726N
0940549E
100
04/22
2286 X 45
AAI
62
Lucknow (Amousi)
Uttar Pradesh
264543N
0805300E
123.14
09/27
2742 X 45
AAI
63
Ludhiana
Punjab
305120N
0755706E
254
12/30
1463 X 30
AAI
64
Madurai
Tamil Nadu
095007N
0780518E

140
09/27
2285 X 45
AAI
65
Malda
West Bengal
250040N
880750E
24
11/29
1099 X 30
AAI
66
Mangalore (BAJPE)
Karnataka
125743N
0745323E
102.6
06/24
2450 X 45
AAI
67
Mumbai (CSI Airport)
Maharashtra
190530N
0725158E
11.9
09/27
14/32
3448 X 60
2871 X 45
AAI Airport Leased to MIAL
68
Muzzafarpur
Bihar
260701N
0851854E
53
11/29
1219 X 30
AAI
69
Mysore
Karnataka
121345N
0763930E
716
05/23
09/27
1350 X 30

1740 X 30
AAI
70
Nadirgul
Andra Pradesh
171617.8N
0783236.2E
552
14/32
914 X 23
AAI
71
Nagpur (Sonegaon) (Mihan)
Maharashtra
210531N
0790254E
314.85
14/32
3200 X 45
AAI airport operated by MIPL
72
Panna
Madhya Pradesh
243915N
801540E
424
17/35
1538 X 18
AAI
73
Pantnagar
Uttarakhand
290156N
0792821E
233
10/28
1372 X 30
AAI
74
Passighat
Arunachal Pradesh
2806N
9523E
157
17/35
1006 X 18
AAI
75
Patna
Bihar
253537N

0850531E
51.18
07/25
2072 X 45
AAI
76
Pondicherry
Pondicherry
115759N
0794843E
43
07/25
1502 X 30
AAI
77
Porbandar
Gujarat
213901N
0693931E
7
09/27
1372 X 45
AAI
78
Raipur (MANA)
Chattisgarh
211052N
0814419E
317.30
06/24
2286 X 45
AAI
79
Rajahmundry
Andhra Pradesh
170631N
0814918E
45
05/23
1750 X 45
AAI
80
Rajkot
Gujarat
221834N
0704646E
134.4
05/23
1846 X 45
AAI
81

Ranchi (Birsa Munda Airport)

Jharkhand

231851N

0851916E

654.71

13/31

2713 X 45

AAI

82

Raxaul

Bihar

26 59 48N

84 49 14E

79

10/28

1097 X 30

AAI

83

Rupsi

Assam

2608N

8945E

40

05/23

1829 X 45

AAI

84

Safdarjung Airport

Delhi

283500N

0771229E

212

12/30

1180 X 45

AAI

85

Salem

Tamil Nadu

114647N

0780355E

300

04/22

1829 X 45

AAI

86

Satna

Madhya Pradesh

2434N

08051E

315

11/29

1829 X 30
AAI
87
Shimla (Jubbarhatti)
Himachal Pradesh
310454N
0770407E
1540
14/32
1189 X 23
AAI
88
Sholapur
Maharashtra
173735N
0755606E
481
15/33
1365 X 45
AAI
89
Surat
Gujarat
210647N
0724435E
6
04/22
2250 X 45
AAI
90
Tirupathi
Andhra Pradesh
133759N
0793231E
106.75
08/26
2286 X 45
AAI
91
Tiruchirapalli (Trichy)
Tamil Nadu
104556N
0784254E
87.78
09/27
2427 X 4
AAI
92
Thiruvananthapuram
Kerala
082847N

0765511E

4

14/32

3398 X 45

AAI

93

Tezu

Arunchal Pradesh

27 54N,

96 04E

220

04/22

1372 X 30

AAI

94

Tuticorin

Tamil Nadu

084317N

780141E

26

10/28

1350 × 30

AAI

95

Udaipur (Maharana Pratap Airport)

Rajasthan

243703N

0735340E

513.28

08/26

2281 X 45

AAI

96

Vadodara

Gujarat

221948N

0731308E

39.32

04/22

2469 X 45

AAI

97

Varanasi (Babatpur)

U.P.

252705N

0825131E

81

09/27

2745 X 45

AAI

98

Vellore

Tamil Nadu

125424N

0790406E

233

07/25

793 X 150

AAI

99

Vijayawada

Andhra Pradesh

163102N

0804812E

25

08/26

2286 X 45

AAI

100

Warangal

Andhra Pradesh

175452N

0793608E

284

09/27

1859 X 45

AAI

IV

Part-1 Aerodromes Operated By The State Governments & Private Owners

(Controlled Or Public Use Aerodromes)

S.No.

Airport

State

Coordinates

Aerodrome Elevation in Meters

Runway

Dimension in Meters

Owner/ Operator

(1)

(2)

(3)

(4)

(5)

(6)

(7)

1

Aizawl (Lengpui)

Mizoram

235016.88N

0923736.38E

418

17/35

2500x45
SG
2
Diu
Union Territory
204247N
0705514E
4.9
05/23
13/31
1845 x 45
1069 x 25
U.T.
3
Durgapur
West Bengal
233727.7N
0871432.5E
85
16/34
3315x45
BAPL
4
Puttaparthi
Andhra Pradesh
140853N
0774726E
478.23
09/27
2224x45
PVT
5
Vijaynagar
Karnataka
151019N
0763837E
502
13/31
1600 x 30
PVT
6
Mundra
Gujarat
225003N
0694552E
5.18
05/23
1700X30
PVT
7
Jamshedpur

Jharkhand
22 48 46.71N
86 10 10.76E
141.7
08/26
1220X45
PVT
8
Baramati
Maharashtra
181335.84N
743522.91E
605
11/29
1172X30
PVT
9
Beas
Punjab
313332N
0752000E
233
16/34
2552X45
PVT
10
Latur
Maharashtra
182437.9N
0762752.9E
651
05/23
1700X30
PVT
11
Nanded
Maharashtra
191051.73N
0771921.17E
379
10/28
2300X45
PVT
12
Osmanabad
Maharashtra
181643.55N
0760317.16E
689
04/22
1190X30

PVT

13

Yavatmal

Maharashtra

202344.27N

0781226.32E

429

08/26

1218X30

PVT

IV

Part-2 Aerodromes Operated By The State Governments & Private Owners

(Uncontrolled Or Private Use Aerodromes)

S. No.

Airport

State

Coordinates

Aerodrome Elevation in Meters

Runway

Dimension in Meters

Owner/ Operator

(1)

(2)

(3)

(4)

(5)

(6)

(7)

1.

Abu Road

Rajasthan

242940N

0724652E

255

09/27

1219 X 23

S.G.

2.

Akbarpur

U.P.

2627N

08234E

101

11/29

1829 X 45

S.G.

3.

Aligarh

U.P.

275140N

0780847E

NA
11/29
1097 X 23
S.G.
4.
Ambari
West Bengal
26 34 30N
88 32 30E
107
18/36
1219 X 24
PVT
5.
Ambikapur (Darima)
Chattisgarh
2259N
08312E
579
16/34
1372 X 15
S.G.
6.
Amla
M.P.
2156N
78078E
746
08/26
1067 X 30
S.G.
7.
Amravati
Maharashtra
20 48 48N
0774303E
341
08/26
1372 X 30
S.G.
8.
Amreli
Gujarat
2137N
07113E
137
13/31
914 X 45
S.G.
9.
Baldota Koppal

Karnataka

152137N

0761253E

522

10/28

1600 X 45

PVT

10.

Banasthali

Rajasthan

26 24 26N

0755209E

308

09/27

1357 X 60

PVT

11.

Bangalore (IIS)

Karnataka

13 01 25N

77 34 13E

933

09/27

655 X 30

PVT.

12.

Banswara (Tilwara)

Rajasthan

23 35 22N

0741841E

180

10/28

1250 X 15

S.G.

13.

Basant Nagar

Andhra Pradesh

1842N

07924E

204

09/27

1529 X 45

PVT.

14.

Betul

M.P.

2152N

7758E

549

08/26

914 X 45

S.G.
15.
Bhagalpur
Bihar
2515N
08701E
45
09/27
1067 X 30
S.G
16.
Bhilai (Nandani)
Chattisgarh
2118N
08123E
302
05/23
1524 X 30
PVT.
17.
Bhiwani
Haryana
2850N
07611E
217
12/30
1088 X 23
S.G.
18.
B.H.U. Flying Club
U.P.
251513.5N
82 5926.3E
83M
08/26
695 X 45
BHU
19.
Birlagram (Nagda)
M.P.
2327N
07525E
469
05/23
1463 X 30
PVT.
20.
Birpur
Bihar
263034N
0870104E

74
09/27
1097 X 90
S.G.
21.
Birwa
M.P
22 05N
80 35E
552
14/32
1400 X 24
S.G
22.
Bokaro
Jharkhand
233827N
0860853N
216
13/31
1400 X 45
PVT.
23.
Borengajuli
Assam
2645N
09149E
122
02/20
1020 X 90
PVT.
24.
Burhar (Shahdol)
M.P.
231400N
813000E
480
14/32
1224 X 15
PVT.
25.
Burnpur
West Bengal
233751N
0865830N
94
E/W
914 X 90
PVT.
26.
Begusarai

Bihar
2525N
08605E
41
09/27
762 X 90
S.G.
27.
Berhampur (Korapalli)
Orrisa
191753N
845237E
37
18/36
750 X 15
S.G.
28.
Chandrapur
Maharashtra
19 59 42N
079 13 18E
244
08/26
1000 X 30
S.G.
29.
Chhindwara
M.P.
2200N
07855E
665
11/29
1486 X 30
S.G.
30.
Chillari
Kerala
1107N
07553E
770
10/28
1045 X 15
PVT.
31.
Chinyali Saur
Uttarakhand
30 34 59.90N
078 19 22.47E
854
16/34
1000 X 23

S.G.
32.
Cuttack (Charbatia)
Orissa
2033N
08554E
41
04/22
03/31
2286 X 45
1465 X 45
ARC
33.
Chetnad
Tamil Nadu
1010N
07848E
107
06/24
14/32
1829 X 45
1463 X 45
S.G.
34.
Chhapra
Bihar
2547N
08446E
53
L/A
914 X 45
S.G.
35.
Daltonganj
Jharkhand
24 01 13N
084 05 46E
335
09/27
914 X 45
S.G.
36.
Damoh
M.P.
2402N
07925E
348
06/24
1524 X 76
PVT.
37.

Deoghar
Jharkhand
24 26 47N
086 42 26E
110
L/A
731 X 45
SG
38.
Dhanbad
Jharkhand
2350N
08626E
260
09/27
1128 X 23
SG
39.
Dhana
M.P.
234507.3N
785142.3E
529
18/36
823 X 41
S.G.
40.
Dhulia
Maharashtra
2055N
07444E
289
05/23
09/27
1828 X 45
1372 X 30
S.G.
41.
Dumka
Jharkhand
24 13 52N
87 16 12E
137
09/27
640 X 90
S.G.
42.
Durgapur Steel Plant
West Bengal
23 34 58N
87 20 25E

88M
12/30
1200 X 23
NA
43.
Etawah (SAIFAI)
U.P.
2636 01N
79 03 35E
NA
15/33
1704 X 23
S.G.
44.
Faizabad
U.P.
26 45 00N
082 09 17E
314
11/29
05/23
1815 X 45
1429 X 45
S.G.
45.
Fursatgan
U.P.
2615N
08123E
108
09/27
1722 X 45
IGRUA
46.
Gaucher
Uttarakhand
301729N
790850E
740
12/30
1200 X 23
S.G.
47.
Ghazipur
U.P.
2537N
08334E
68.5
07/25
1807 X 45
S.G.

48.
Guna
M.P.
2439N
07721E
495
14/32
914 X 23
S.G.
49.
Hamirgarh
Rajasthan
2508N
07437E
419
18/36
1274 X 30
S.G.
50.
Hadapsar (Gliderome)
Maharashtra
18 29 32N
073 56 26E
579
E/W
1052 X 121
SG
51.
Hosur
Karnataka
12 39 44N
77 46 12E
930
09/27
1219 X 30
PVT.
52.
Hirakund
Orissa
2135N
08400E
208
15/33
1097 X 45
S.G.
53.
Hisar
Haryana
2911N
07546E
214

12/30
1219 X 45
S.G.
54.
Jagdalpur
Chhattisgarh
1904N
08202E
547
06/24
1125 X 30
S.G.
55.
Jakkur
Karnataka
130432N
0773546E
919
08/26
854 X 21
S.G.
56.
Jashpurna Gar
Chhattisgarh
225558N
0841341E
457
09/27
1067 X 23
S.G.
57.
Jaypore
Orissa
1853N
08233E
595
16/34
916 X 30
S.G.
58.
Jhabua (Ranpet)
M.P.
22 46N
74 33E
435
09/27
792 X 30
S.G.
59.
Jhingura
U.P.

2508N
08239E
91
09/27
1220 X 45
S.G.
60.
Jhunjhunu
Rajasthan
280620N
752240E
335
10/28
1014 X 15
S.G.
61.
Kanpur (Kalyanpur)
U.P.
263113N
801357E
131
09/27
884 X 23
PVT.
62.
Kankroli
Rajasthan
250114N
735359E
532.46
12/30
1100 X 23
PVT.
63.
Karad
Maharashtra
1717N
07409E
576
09/27
1280 X 30
S.G.
64.
Kargil
J & K
343133N
0760924E
2920
02/20
1829 X 30
SG

65.
Karnal
Haryana
2942N
07702E
246
13/31
1170 X 30
S.G.
66.
Kasia (Kushinagar)
U.P.
264612N
835429E
76
11/29
1722 X 23
S.G.
67.
Kayattar
Tamil Nadu
08 58 12.85N
77 49 12.57E
91
09/27
06/24
1463 X 45
1829 X 30
S.G.
68.
Kolapni
Assam
26 48 07N
93 12 45E
90
04/22
914 X 90
PVT.
69.
Khargone
M.P
2148N
07533E
276
09/27
1000 X 24
S.G.
70.
Kishanganj
Bihar
260433N

875623E
148
09/27
1006 X 90
S.G.
71.
Lonavala (Amby Valley)
Maharashtra
183634N
732242E
689
14/32
1199 X 30
PVT.
72.
Lalgarh
Rajasthan
295100N
740100E
176
L/A
1005 X 174
S.G.
73.
Mandla
M.P
22 30N
80 20E
489
09/27
1800 X 36
S.G
74.
Madhaiganj
West Bengal
23 38 30N
87 20 37E
NA
15/33
05/23
1800 X 50
1400 X 50
NA
75.
Mandvi
Gujarat
22 50 22N
69 18 17E
5
08/26
1400 X 30

S.G.
76.
Mackebpur
Assam
2651N
09446E
67
14/32
1005 X 91
PVT.
77.
Meerut
U.P.
2854N
07741E
NA
11/29
1829 X 23
S.G.
78.
Mithapur (Dwarka)
Gujarat
222440N
685934E
3.6
07/25
1372 X 45
914 X 45
PVT.
79.
Muirpur (Korba)
U.P.
240729N
830217E
405
09/27
823 X 60
PVT.
80.
Madhubani
Bihar
261945N
860338E
47
18/36
914 X 45
S.G.
81.
Mantalai
J & K
33 00 13N

75 21 21E
3389
15/33
488 X 30
PVT.
82.
Mathania
Rajasthan
26 26N
073 06E
251
05/23
1737 X 45
S.G.
83.
Mehsana
Gujarat
233612N
0722230E
85
05/23
914 X 45
S.G.
84.
Munger
Bihar
25 20 47N
86 28 59E
41
L/A
732 X 91
S.G.
85.
Nagaur
Rajasthan
2712N
07343E
281
05/23
1170 X 30
S.G.
86.
Nagda (Birlagram)
M.P.
2327N
7524E
366
13/31
1463 X 30
PVT.
87.

Naini/ Saini Pithoragarh
Uttarakhand
293539N
801425E
1463
14/32
1330 X 20
S.G.
88.
Narnaul
Haryana
2805N
07612E
295
09/27
914 X 23
S.G.
89.
Neemuch
M.P.
2425N
07452E
497
14/32
1700 X 30
CRPF
90.
New Lands
West Bengal
2639N
08948E
122
11/29
1076 X 90
PVT.
91.
Neyveli
Tamil Nadu
1137N
07932E
53
05/23
914 X 30
PVT.
92.
Nagarjuna Sagar
Andhra Pradesh
1632N
07919E
259
NA

1654 X 30

S.G.

93.

Naria See Saharsa

Bihar

25 53 32N

86 35 00E

40

L/A

457 X 91

S.G.

94.

Nawapara

Orissa

2052N

08230E

322

05/23

1002 X 18

S.G.

95.

Ondal Airport

West Bengal

23 37 19N

83 14 35E

82

14/32

04/22

1829 X 45

1463 X 45

S.G.

96.

Pachmarhi

M.P.

2227N

07824E

1085

04/22

1200 X 61

S.G.

97.

Panneri

Assam

2643N

09154E

91

18/36

1060 X 90

PVT.

98.

Patiala

Punjab
3019N
07622E
250
15/33
1097 X 45
S.G.
99.
Pilani
Rajasthan
282100N
753500E
335
05/23
914 X 45
PVT.
100.
Pinjore
Haryana
304926N
765258E
500
16/34
914 X 45
S.G.
101.
Pirthiganj
U.P.
2552N
08201E
94
12/30
1829 X 45
S.G.
102.
Pithoragarh
Uttarakhand
2940N
08013E
1463
14/32
1300 X 23
S.G.
103.
Padampur
Orissa
2102N
08303E
198
18/36
914 X 54

S.G.
104.
Raigarh (Jindal Airstrip)
Chattisgarh
215613N
832044E
242
10/28
2000 X 30
PVT.
105.
Raipur (Baikunth)
Chattisgarh
21 29 54N
81 47 37E
307
06/24
1353 X 24
PVT.
106.
Rajhara (Dhalli)
Chattisgarh
20 31 40N
81 04 57E
361
05/23
914 X 45
PVT.
107.
Rewa
M.P.
2430N
08113E
305
06/24
1200 X 30
S.G.
108.
Rourkela
Orissa
2216N
08449E
210
09/27
1615 X 30
PVT.
109.
Ratlam
M.P.
2322N
7501E

517
08/26
1200 X 23
S.G.
110.
Sagar
M.P
23 45N
78 51E
574
18/36
995 X 23
S.G
111.
Sedam
Karnataka
171000N
771800E
427
05/23
488 X 30
PVT.
112.
Seoni
M.P
21 56N
79 30E
633
10/28
1800 X 30
S.G
113.
Shahdol
M.P.
2314N
08130E
480
14/32
1224 X 15
S.G.
114.
Shivpuri (Burhar)
M.P.
2524N
07740E
396
09/27
913 X 23
BSF
115.
Shravasti

U.P.
27 30N
82 02E
NA
12/30
1829 X 23
S.G.
116.
Sidhi
M.P.
2424N
08149E
366
06/24
1006 X 15
S.G.
117.
Sirohi
Rajasthan
24 53 32N
72 51 0.9E
297.5
13/31
1830 X 18
S.G.
118.
Sitamau
M.P.
24 01 10N
75 20 17E
479
L/A
823 X 45
S.G.
119.
Sultanpur (Amhai)
U.P.
2615N
08202E
91
11/29
1829 X 45
S.G.
120.
Saharsa
Bihar
255334N
863510E
45
09/27
914 X 45

S.G.
121.
Sawai Madhopur
Rajasthan
260200N
762100E
266
18/36
914 X 45
S.G.
122.
Tekanpur
M.P.
2600N
07816E
213
08/26
1311 X 46
BSF
123.
Tura
Meghalaya
253942N
902041E
534
16/34
1005 X 30
SG
124.
Tushra
Orissa
203043N
832653E
168
06/24
1214 X 45
S.G.
125.
Ujjain
M.P.
2306N
07553E
543
06/24
1219 X 22
S.G.
126.
Umaria
M.P
23 32N
80 48E

451
17/35
1500 X 25
S.G
127.
Utkela
Orissa
2006N
08311E
229
04/22
914 X 45
S.G.
128.
Uttarkashi
Uttarakhand
7819N
08035
853
16/34
NA
S.G.
129.
Yinghiong
Arunachal Pradesh
28 38 20N
95 01 10E
500
NA
975 X 18
S.G.
130.
Ziro
Arunachal Pradesh
27 35 17N
93 49 42E
1524
18/36
1219 X 30
SG
V
Defence Aerodrome
S.No.
Airport
State
Coordinates
Aerodrome Elevation in Meters
Runway
Dimension in Meters
Owner/ Operator
(1)

(2)

(3)

(4)

(5)

(6)

(7)

1.

Adampur

Punjab

312616N

754526E

247

13/31

2746 X 45

IAF

2.

Agra

Uttar Pradesh

270932N

775730E

167.7

05/23

2744 X 45

IAF

3.

Allahabad

Uttar Pradesh

252626N

814409E

97.2

12/30

2477 X 45

IAF

4.

Ambala

Haryana

302219N

764850E

275

12/30

2815.7 X 46

IAF

5.

Arrakonam

Tamil Nadu

130358.73N

794059.98E

85

06/24

4095 X 46

Indian Navy

6.
Awantipur
Jammu & Kashmir
335239N
745836E
1647
12/30
3200 X 45
IAF

7.
Bagdogra (Siliguri)
West Bengal
264108N
0881948E
126
18/36
2744 X 45
IAF

8.
Bakshi-Katalab
Uttar Pradesh
265914N
805336E
124
09/27
2743 X 46
IAF

9.
Bareilly
Uttar Pradesh
282519N
792705E
173
11/29
2743 X 46
IAF

10.
Bareilly
Uttar Pradesh
282000N
792705E
168
12/30
900 X 45
Army

11.
Barrackpore
West Bengal
224656N
882146E
06

02/20
1866 X 45
IAF
12.
Bengaluru (HAL)
Karnataka
125703N
0773957E
888
09/27
3306 X 45
HAL
13.
Bhatinda
Punjab
301611N
744523E
203
13/31
2805 X 46
IAF
14.
Bhatinda
Punjab
301358.1N
750255.8E

09/27
13/31
650 X 15
1200 X 50
Army
15.
Bhuj (Rudramata)
Gujarat
231713N
0694015E
78
05/23
2515 X 45
IAF
16.
Bidar
Karnataka
175429N
772900E
664
08/26
2700 X 45
IAF
17.

Bihta
Bihar
253524N
845306E
54
10/28
2210 X 46
IAF
18.
Bikram Park
Jammu & Kashmir
325400
750600E
621
03/21
1148 X 85
Army
19.
Campbell Bay
Andaman & Nicobar Islands
070046.75N
0935524.50E
1
05/23
1050 X 30
Indian Navy
20.
Car Nicobar
Andaman Islands
090915N
924913E
10.8
02/20
2713 X 46
IAF
21.
Chabua
Assam
272742N
950714E
110
05/23
2746 X 46
IAF
22.
Chandigarh
Union Territory
3040N
07647E
314
11/29

2744 X 45
IAF
23.
Charbatia
Orissa
2033N
08554E
41
04/22
03/31
2286 X 45
1465 X 45
ARC
24.
Daman
Union Territory
202559.18N
0725035.22E
12.8
03/21
10/28
1801 X 45
1001 X 25
I.C.G
25.
Darbhanga
Bihar
261137N
855503E
47
10/28
2743 X 46
IAF
26.
Dinjan
Assam
273243.1N
941419.7E
120
07/25
575 X 35
Army
27.
Dundigal
Andhra Pradesh
173740N
782411E
613
10/28
2513 X 46
IAF

28.
Goa (Dabolim)
Goa
152247.41N
0734940.05E
46
08/26
3430 X 45
Navy
29.
Gorakhpur
Uttar Pradesh
264429N
0832709E
78
11/29
2744 X 45
IAF
30.
Gwalior
Madhya Pradesh
261730N
0781341E
188
06/24
2744 X 45
IAF
31.
Hakimpet
Andhra Pradesh
173308N
783133E
613
09/27
2110 X 46
IAF
32.
Halwara
Punjab
304457N
753751E
239
13/31
2743 X 46
IAF
33.
Haldwani
Uttarakhand
291452N
793239E
478

01/19
150 X 50
Army
34.
Hasimara
West Bengal
264150N
892217E
109
11/29
2744 X 46
IAF
35.
Hindon
Uttar Pradesh
284223N
772138E
214
09/27
2743 X 46
IAF
36.
Jaisalmer
Rajasthan
265323N
0705200E
236
04/22
2744 X 45
IAF
37.
Jalandhar
Punjab
311750N
753651E
234
14/32
600 X 45
Army
38.
Jammu
Jammu & Kashmir
324119N
745017E
291
18/36
2042 X 45
IAF/CE
39.
Jamnagar
Gujarat

222800N
700100E
15
06/24
2514 X 46
IAF
40.
Jhansi
U.P.
252935.4N
783366.9E

15/33
1070 X 20
Army
41.
Jodhpur
Rajasthan
261508N
730300E
217
05/23
2743 X 45
IAF
42.
Jorhat
Assam
2644N
09411E
91
04/22
2652 X 45
IAF
43.
Kalaikunda
West Bengal
222012N
871307E
61
17/35
2742 X 46
IAF
44.
Kanpur (Chakeri)
Uttar Pradesh
2624N
08025E
124
09/27
2744 X 45
IAF

45.
Khalsi
Jammu & Kashmir
341900N
765400E
3005
29/11
245.2 X 27.8
Army
46.
Kochi
Kerala
095638.25N
0761623.89E
2
17/35
13/31
1830 X 46
1416 X 46
Indian Navy
47.
LEH
Jammu & Kashmir
3408N
07733E
3256
07/25
2922 X 45
IAF
48.
LEH
Jammu & Kashmir
3407N
07733E
3256
06/24
400 X 26
Army
49.
Leimakhong
Manipur
245672N
935113E
841.55
01/19
350 X 50
Army
50.
Mamun Cantt
Punjab
321700N

754300E
397
18/36
3234 X 83
Army
51.
Manasbal
Jammu & Kashmir
341456N
743855E
1594
17/35
1100 X 50
Army
52.
Missamari
Assam
264901N
923551.5E
95
05/23
1521 X 45.72
Army
53.
Nagrota
Jammu & Kashmir
324700N
7543855E
347
02/20
132 X 15
Army
54.
Nagtalao
Rajasthan
2625N
7307E
240
02/20
1400
Army
55.
Nal (Bikaner)
Rajasthan
280418N
731225E
215
05/23
2746 X 45
IAF
56.

Naliya
Gujarat
231321N
685329E
42
06/24
2743 X 46
IAF
57.
Nasik
Maharashtra
195744.32N
734831.74E
599
09/27
1373 X 46
Army
58.
NDA
Maharashtra
182822N
734646E
610
09/27
900 X 15
NDA
59.
Ozar (Nasik)
Maharashtra
2007N
07355E
598
09/27
3000 X 45
HAL
60.
Partapur
Jammu & Kashmir
3456N
7726E
3081
13/31
75 X 25
Army
61.
Pathankot
Punjab
321402N
0753802E
312
01/19

2744 X 45

IAF

62.

Panagarh

West Bengal

232824N

87 25 46E

73

15/33

2544 X 46

IAF

63.

Phalodi

Rajasthan

270618N

721257E

244

05/23

3050

IAF

64.

Patiala

Punjab

301855N

762154E

250

15/33

1170

Army

65.

Port Blair (Veer Savarkar Airport)

Andaman Islands

113854N

924406E

5

04/22

3414 X 45

Indian Navy

66.

Pune (Lohegaon)

Maharashtra

183458N

0735513E

592

10/28

2535 X 45

IAF

67.

Purnea

Bihar

254543N

872442E
37
09/27
2743 X 46
IAF
68.
Ramnd
Tamil Nadu
091910.30N
0785823.40E
4
01/19
3017 X 30
Indian Navy
69.
Ranchi
Jharkhand
231851.3N
851915.8E
-
14/32
263 X 63
Army
70.
Rangapahar
Nagaland
255118N
934220E
182
06/24
238 X 25
Army
71.
Ratnagiri
Maharashtra
170048N
0731944E
92.8
05/23
1800 X 45
I.C.G
72.
Sarsawa (Saharanpur)
Uttar Pradesh
295942N
772532E
271
09/27
2744 X 46
IAF
73.

Sevoke Road
West Bengal
264700N
882700E
144
16/34
883 X 45
Army
74.
Shibpur
Andaman & Nicobar Islands
131408.70N
0930259.00E
3
18/36
1000 X 30
Indian Navy
75.
Sharifabad
Jammu & Kashmir
340500N
744300E
1583
09/27
430
Army
76.
Silchar (Khumbigram)
Assam
245448N
0925851E
103
06/24
1785 X 45
IAF
77.
Sisa
Haryana
293335N
750027E
199
05/23
2743 X 46
IAF
78.
Srinagar
Jammu & Kashmir
3359N
07447E
1656.5
13/31

3658 X 45
IAF
79.
Sulur
Tamil Nadu
110047N
770945E
381
05/23
2520 X 46
IAF
80.
Suratgarh
Rajasthan
292317N
735415E
173
05/23
2743 X 46
IAF
81.
Tambaram
Tamil Nadu
125419N
800719E
27
05/23
12/30
1514 X 46
1815 X 46
IAF
82.
Tezpur
Assam
2643N
09247E
70
05/23
2744 X 45
IAF
83.
Thanjavur
Tamil Nadu
104311N
790610E
76
07/25
14/32
1833 X 45
1465
IAF

84.
Thoise
Jammu & Kashmir
343914N
772237E
3065
11/29
3050 X 46
IAF
85.
Udhampur
Jammu & Kashmir
325411N
750920E
634
18/36
2754 X 46
IAF
86.
Uttarlai
Rajasthan
254841N
712859E
154
02/20
2743 X 46
IAF
87.
Yelahanka
Karnataka
130806N
773636E
928.6
09/27
2440 X 45
IAF
88.
Vishakapatnam
Andhra Pradesh
174316N
0831329E
3
05/23
10/28
1829 X 45
3048 X 45
Navy
VI
Communication, Navigation and Surveillance (CNS) Facilities Located away from the Airports
Sl. No.
Station

Name of the facility
Coordinates in WGS 84
Facility Reduced Level (Meter)
Airport responsible to Safeguard the facility
North
East
1
Aligarh
CVOR/ DME(HP)
27°49' 45.5"
78° 10' 42"
205
Delhi IGI
2
Behrampur
MSSR
19°20' 3.4"
84° 52' 0.2"
37.8
Bhubaneshwar
3
Bellary
MSSR
15° 09' 59.36"
76° 53' 2.01"
489
Bangalore (AAI)
4
Bellary
DVOR/ DME(HP)
15° 9'54.4"
76° 52' 50"
461
Bangalore (AAI)
5
Bikaner-Lunka 1
LUN DVOR/ DME(HP)
28° 33' 09.84"
73° 47' 15.61"
201.77
Jaipur
6
Bikaner-Lunka 2
LKA DVOR/ DME(HP)
28°11' 20.42"
74° 06' 41.267"
255.42
Jaipur
7
Chillarki
CVOR/ DME(HP)

28° 20' 51.2"
76° 39' 57.7"
228.6
Delhi IGI
8
Daman 'DMN'
DVOR/ DME
20 26' 32.52"
72 51' 15.95"
12.8
Daman (coastguard)
9
Daman 'DM'
NDB
20 27' 11.23"
72 51' 09.24"
12.8
Daman (coastguard)
10
Gulbarga
DVOR/ DME(HP)
17° 18' 48.8"
76° 48' 11"
454
Bangalore (AAI)
11
Jalalabad
CVOR/ DME(HP)
27° 41' 39.6"
79° 39' 44.7"
55.5
Delhi IGI
12
Jharsuguda
NDB
21° 53' 47.6"
84° 02' 19.5"
237
Jharsuguda
13
Jharsuguda
MSSR
21° 54' 31.60"
84° 03' 50.40"
262
Jharsuguda
14
Kanchipuram
DVOR/ DME(HP)
12° 47' 5.8"
79° 42' 47.2"

82.89
Chennai
15
Katihar
DVOR/ DME
25° 36' 52.65"
87° 33' 19.20"
31
Kolkata
16
Katihar
MSSR
25° 34' 23.01
87° 33' 20.19"
59
Kolkata
17
Khammampet
NDB
17° 15' 45"
80° 08' 15"
116.5
Hyderabad
18
Pratapgarh
DVOR/ DME(HP)
24° 02' 13.65
74° 44' 38.16"
500
Jaipur
19
Rajamundri
NDB
17° 06' 31"
81° 49' 18"
42.6
Rajahmundry
20
Sakras
CVOR/ DME(HP)
27° 50' 54.4"
77° 00' 29.6"
116.2
Delhi IGI
21
Sampla
CVOR/ DME(HP)
28° 49' 11.1"
76° 49' 9.6"
235.9
Delhi IGI

22

Sikandrabad

CVOR/ DME(HP)

28° 23' 36.2"

77° 42' 29.2"

203.25

Delhi IGI

23

Songarh

NDB

21° 10' 2.5"

73° 33' 57.4"

121.9

Ahmedabad

24

Tuticorin

NDB

08° 43' 32.2"

78° 01' 32.5"

25.9

Chennai

25

Vikarabad

NDB

17° 20' 3.8"

77° 53' 55.5"

651.5

Hyderabad

26

Cochin-II

DVOR "CIB"

10° 07' 05.7"

76° 40' 42.7"

41.3

Cochin

27

Portblair

DVOR "PPB"

11° 38' 58"

92° 44' 50"

154.43

Port Blair

28

Vizag

DVOR "VVZ"

17° 40' 08.6"

83° 15' 11.8"

337.23

Vizag

29

Vizag (Arada Hill)

MSSR

17° 40' 20.0"

83° 15' 37.8"

322

Vizag

30

Chennai (PURUR)

MSSR

13° 01' 46.42"

80° 09' 20.12"

41

Chennai

31

Kolkata (BADU)

MSSR

22° 41' 22.76"

88° 29' 06.05"

27

Kolkata

32

Porbander

MSSR

21° 38' 28"

69° 39' 45.00"

32

Porbandar

VII

Greenfield Airports

For Which Government Of India Has Given

"In-Principle" Approval

No.

Airport(1)

State(2)

Coordinates(3)

Aerodrome Elevation In Meters(4)

Runway(5)

Dimension In Meters(6)

Owner/ Operator(7)

1.

Dabra

Madhya Pradesh

254942.8N

078191.7E

240

09/27

3000 x 45

M/s Gwalior Agriculture Company Ltd.

2.

Navi Mumbai Internationa Airport

Maharashtra

18 59 39.78N

073 30 12.95E

8.00

08L/26R

08R/26L

3700 x 60

3700 x 60

CIDCO

3.

Pakyong

Sikkim

271358.269N

0883518.7927E

1321.27

02/20

1700 x 30

AAI

4.

Mopa

GOA

Master Plan of the airport is Yet to be finalized

State Government

5.

Kannur

Kerala

11° 54' 56.633"N

75° 32' 44.604"E

105

07/25

3400 x 45

KIAL

6.

Sindhudurg

Maharashtra

16° 00' 12.17"N

73° 31' 57.9"E

64

09/27

3045 x 60

MIDC

7.

Bijapur

Karnataka

Master Plan of the airport is Yet to be finalized

State Government

8.

Hassan

Karnataka

Master Plan of the airport is Yet to be finalized

State Government

9.

Gulbarga

Karnataka

Master Plan of the airport is Yet to be finalized

State Government

10.

Simoga

Karnataka

Master Plan of the airport is Yet to be finalized

State Government

11.

Kushinagar International Airport

Uttar Pradesh

Master Plan of the airport is Yet to be finalized

State Government

12.

Karaikal Airport

Puduchery

Master Plan of the airport is Yet to be finalized

State Government

13.

Shirdi

Maharashtra

19° 41'27.332"N

74° 22' 18.35"E

581.75

09/27

2500x45

MADC

VIII

Appendix - A

Inner Horizontal Surface for runway code 3 & 4

VIII

Appendix - B

VIII

Appendix - C

VIII

Appendix - D

VIII

Appendix - E

VIII

Appendix - F

VIII

Appendix - G

VIII

Appendix - H

ICAO Annex 14 Obstacle Limitation Surfaces (OLS)

VIII

Appendix - I

Inner approach, inner transitional and balked landing obstacle limitation surfaces

VIII

Appendix - J

VIII

Appendix - K

VIII

Appendix - L

Maximum Permissible Heights (in Meters) of Building/ Mast/ Chimney, etc. Based on Annex 14 Criteria

(For Code 3 or 4 Instrument Runway Airports only)

Maximum permissible height may be further restricted due to Annex 10 criteria (owing to various CNS facilities) and also due to DOC 8168 criteria for the protection of PANS-OPS surfaces for different procedures. Site elevation of the site will be subtracted from the permissible top elevation to arrive at maximum permissible height of the building/Mast/Chimneys, etc. Following height table is indicative only and in no way assures the height permissible at a given site.

Table: Maximum Permissible Heights (in Meters)

ICAO Annex 14 Surface (Height in meters)

Distance from Runway Strip (in meters)

500
1000
1500
2000
2500
3000
3500
4000
4500
5000
6000
7000
8000
9000
10000

Approach Surface*

10
20
30
40
45
45
45
48
73
98
120
140
150
150
150

Take-off climb Surface*

10
20
30
40
45
45
45

48
73
98
120
140
150
150
150
Inner Horizontal Surface (IHS)**

45
45
45
45
45
45
45
45
45

-
-
-
-
-
-
-

Conical Surface**

-
-
-
-
-
-
-
-

70
95
145
195
245
295
300

* Runway end elevation should be added to the permissible heights to arrive at the permissible top elevation, AMSL.

** Aerodrome elevation should be added to the permissible heights to arrive at the permissible top elevation, AMSL.

VIII

Appendix - M

List of Designated Officers of Airports Authority of India (AAI)

Airports Authority of India has set up nine NOC processing offices one each at Delhi, Kolkata, Mumbai, Chennai, Guwahati Hyderabad, Bengaluru, Ahmedabad and Nagpur airports these offices are headed by Designated Officers. NOC offices work as per the provisions of this notification and as per the direction of ATMC 2 of 2013, as amended from time to time by AAI.

The list of Designated Officers are as follows:

1. Regional Executive Director, AAI, Northern Region, Delhi Airport, Delhi

2. Regional Executive Director, AAI, Eastern Region, Kolkata Airport, Kolkata
3. Regional Executive Director, AAI, Western Region, Mumbai Airport, Mumbai
4. Regional Executive Director, AAI, Southern Region, Chennai Airport, Chennai
5. Regional Executive Director, AAI, North Eastern Region, Guwahati Airport, Guwahati
6. GM-Coordination In charge, AAI, Hyderabad Airport
7. GM-Coordination In charge, AAI, Bengaluru Airport
8. Airport Director, AAI, Ahmedabad Airport
9. Airport Director, AAI, Nagpur Airport.

IX

Related AI tags, queries and research notes

Translation